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A comprehensive review of AI-driven water-quality monitoring and prediction: advances, challenges, and future directions

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Water quality has emerged as a critical global concern that requires advanced monitoring and management strategies. Traditional water-quality assessment methods predominantly rely on laboratory-oriented analysis that are time consuming, expensive, and are often labor-intensive. This study presents a thorough overview of the paradigm shift from conventional lab analysis towards intelligent and automated, AI-based assessment and monitoring frameworks. The survey systematically explains the use of machine learning (ML), deep learning (DL), and hybrid models to analyze complicated multidimensional and nonlinear water parameters. This study evaluates predictive architectures ranging from traditional regression-based models to advanced ensembles and deep neural networks (CNNs, ANNs, and RNNs) integrated with IoT computing technologies for continuous water quality monitoring. Enhancing models' interpretability and transparency through Explainable Artificial Intelligence methods like SHAP and LIME are analyzed. This survey addresses existing challenges like data scarcity, interpretability of the model, and computational complexity. Self-attention-based architectures, generative AI, and edge computing used to improve the robustness of future water-quality management systems are also assessed. An overall structured research perspective on AI-driven water-quality assessment with methodological gaps with future scope are established in this survey.

KEYWORDS
deep learning, explainable artificial intelligence, generative AI, machine learning, water quality assessment

1 Introduction

Although water covers approximately 71% of the Earth's surface, only about 2.5 to 3.0% of it is freshwater, with an even smaller fraction being potable. Less than 1% is accessible for human use from sources like wells, ponds, shallow groundwater, lakes, and rivers, as a majority of fresh water is locked up as glaciers and deep underground (68.7%). Water-quality assessing is vital for environmental wellness. However, the large volume of data makes traditional and man-monitored assessment approaches challenging, meaning AI would be useful for this complex task. AI helps to automate tasks, create hypotheses, and can produce highly accurate results in a short time, which is beyond human ability (Dilmi and Ladjal, 2021). AI methods have been more effective, cost reductive, and time efficient than conventional assessments,

which are expensive, time consuming, and have a high possibility of error (Nasir et al., 2022). Historical observations are used by Artificial Intelligence to analyze patterns in water-quality prediction models (Huang et al., 2023). Dynamic nonstable, noisy, and nonlinear behavior of conventional environment's data makes it difficult to process water-quality data (Wang Z. et al., 2023). Spatial and temporal dependencies both require highly predictive models which are capable of capturing complexed data (Yang et al., 2021). Machine learning Models offer effective resource management and decision-taking through interactive visualized dashboards (Pandya et al., 2025). Many Hybrid DL models have proven to be effective and superior in evaluating complex relationships in Water Quality Index (WQI) among multiple water quality parameters (Ehteram et al., 2024). Data-scarce environments impact the accuracy of monitoring but can be managed by Self-Attentive models (Huang et al., 2024). Preprocessing methods and normalization techniques on large and irregular datasets highly impact the reliability of DL Models (Mahesh et al., 2024). There are many water-quality analyzing parameters that are essential to determine the quality of water. These are classified into physical, chemical, and biological properties. Physical parameters include Temperature, Turbidity, Total Dissolved Solids, Electrical Conductivity, Color, Taste, and Odor. Chemical Parameters include pH, Dissolved Oxygen, Biological Oxygen Demand, Nutrients, Hardness, and Chemical Pollutants. Biological Parameters include Bacteria, Algae, Viruses, and Aquatic Macroinvertebrates. AI Models enhanced with Feature Selection and Optimization techniques provide more robust and reliable methods (Uddin, 2023; Dheda et al., 2022; Bi et al., 2023). Though AI models work more steadily than traditional approaches, practical implementation is questioned with the trustability of black box models. Explainable Artificial Intelligence (XAI) ensures integrity by analyzing the most effective parameters or indicators (Uddin et al., 2022). Figure 1 demonstrates the comparison of an AI model with a human-guided water-quality prediction model.

1.1 Causes for water quality pollution

Water resources are continually contaminated by environmental and anthropogenic pressures, with industrial effluents and urban runoff representing the primary drivers of degradation (Zhong et al., 2022). Factors that affect water quality are shown in Figure 2.

1.2 Overall AI water quality assessment

The surveyed studies indicate that modern AI-based water quality monitoring frameworks typically begin by acquiring multimodal data from IoT sensors, satellite and UAV imagery, historical laboratory records, and field reports. These pipelines then apply preprocessing workflows, combining dimensionality reduction via Principal Component Analysis (PCA) with feature selection methods such as Pearson correlation and Recursive Feature Elimination (RFE). The literature highlights the application of Machine Learning, Deep Learning, and Generative AI models for WQI prediction, contamination detection, spatiotemporal forecasting, synthetic data generation, and digital twin-based environmental simulation, with federated learning emerging as a promising approach for privacy-preserving distributed model training. Across these studies, model performance is commonly evaluated using regression metrics such as MAE, RMSE, MSE, MAPE, and R^2, as well as classification metrics including Accuracy, Precision, Recall, F1-score, AUC-ROC, and Confusion Matrix. Furthermore, Explainable AI techniques, particularly SHAP and LIME, are increasingly incorporated to improve model interpretability, transparency, and regulatory trust. Overall, the reviewed approaches demonstrate that integrating advanced AI techniques with explainable and privacy-preserving methodologies has significant potential to support real-time water-quality assessment, early contamination detection, treatment optimization, and automated decision support for sustainable water resource management. The

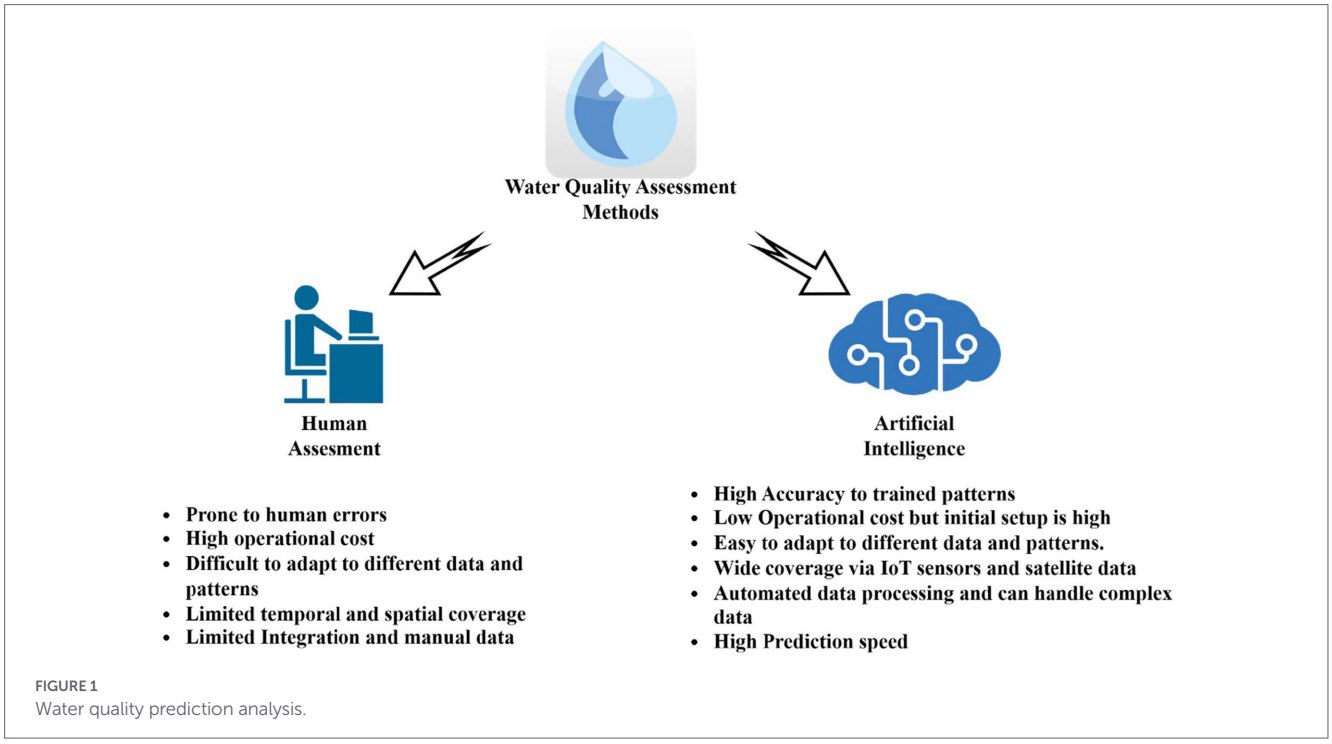


FIGURE 1
Water quality prediction analysis.

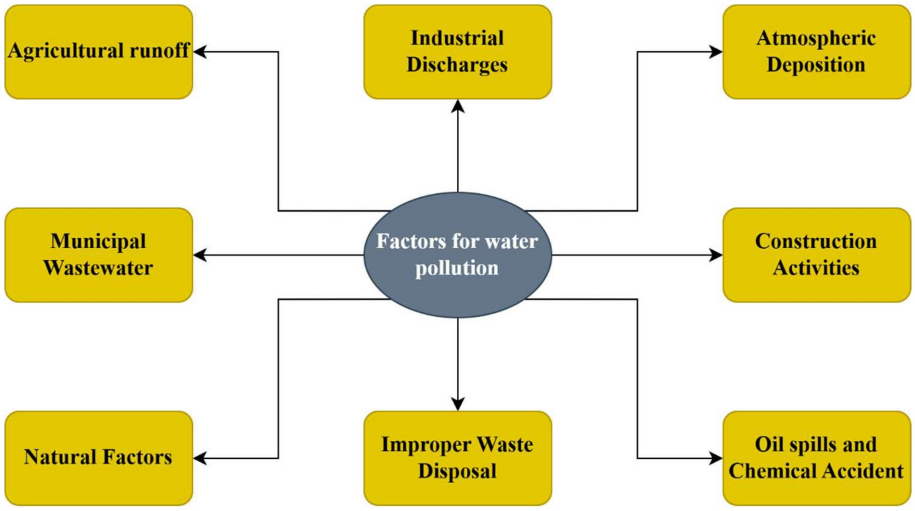


FIGURE 2
Overview of water quality depletion.

conceptual AI framework for water-quality assessment operates across three hierarchical layers, as shown in Figure 3.

1.2.1 Addressing gaps

Despite the rapid growth of AI-driven approaches in water quality research, several critical gaps remain unaddressed in the existing literature.

- Most published studies are conducted in geographically constrained, single-site settings, limiting the generalizability of model architectures to diverse hydrological contexts such as tropical, arid, and glacially influenced watersheds.
- Benchmark comparisons across studies remain inconsistent due to the use of heterogenous datasets, differing input feature sets, and non-standardized evaluation protocols, making it difficult to draw definitive conclusions about model superiority.
- While ensemble and deep learning models demonstrate strong predictive performance, most function as black-box systems that offer limited mechanistic insight into contaminant dynamics, which acts as a critical barrier to regulatory approval and environmental policy integration.
- Temporal dynamics are frequently underexplored: many studies apply ML models to static snapshots of water quality rather than modelling the continuous spatio-temporal evolution of parameters such as dissolved oxygen (DO), turbidity, and nutrient concentrations.
- Data scarcity for rare but high-impact contamination events including industrial chemical spills, food-driven pollutants pulses, and microbial outbreak episodes remain a persistent obstacle that synthetic data generation via Generative AI has only begun to address.
- The operational deployment of AI models on resource-constrained edge devices for rural and low-income regions remains largely unexplored, leaving a gap between laboratory scale model performance and real-world water safety monitoring.

This review directly addresses these gaps by systematically surveying, categorizing, and comparing AI architectures across the full spectrum from classical ML to Generative AI, with emphasis on model interpretability, hybrid design, and future deployment pathways.

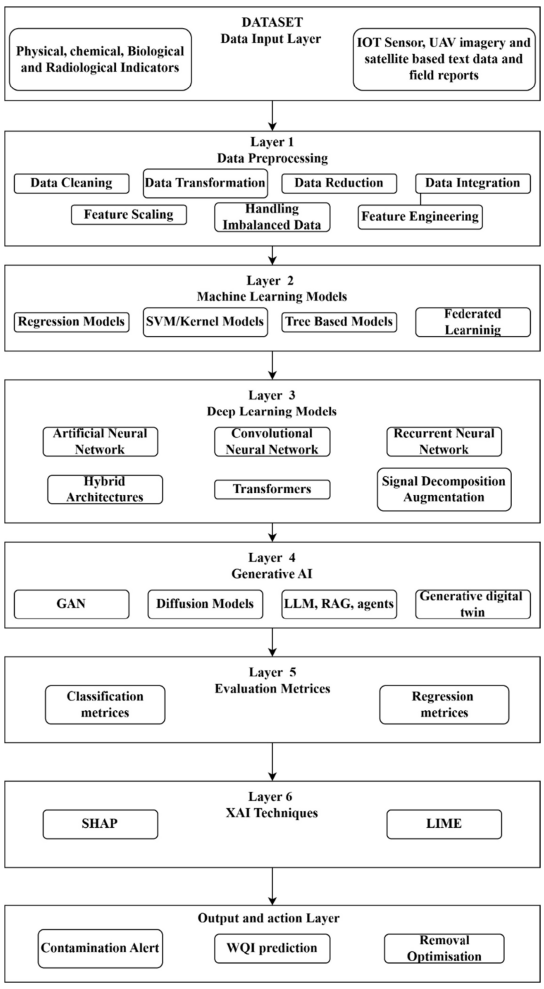


FIGURE 3
Conceptual AI architecture for water-quality assessment.

1.2.2 Main contributions

The principal contributions of this review are as follows:

- Comprehensive Taxonomy of AI Models: This paper provides the first unified, hierarchical taxonomy that classifies water-quality AI methods across three layers, namely classical Machine Learning (Regression, SVM, Ensemble), Deep Learning (ANN, LSTM, GRU, CNN, Transformer, Hybrid architectures), and Generative AI (GANs, Diffusion Models, LLMs), supported by a structured comparative table covering algorithms, task types, strengths, and reported performance metrics.
- Systematic Literature Review via PRISMA: A rigorous PRISMA-guided screening of 351 articles drawn from IEEE Xplore, ScienceDirect, Springer, Nature, Elsevier, and Wiley databases yielded 107 high-quality studies, ensuring reproducibility and minimizing selection bias in evidence synthesis.
- Comparative literature synthesis: The review systematically benchmarks the predictive performance of diverse AI architectures across standardized water-quality parameters including WQI, DO, turbidity, pH, BOD, nitrate, and heavy metal concentrations, providing practitioners with evidence-based model selection guidance.
- Conceptual AI architecture: A conceptual AI framework for water-quality assessment is proposed and illustrated (Figure 3), integrating the data input pipeline, ML/DL inference layers, and Generative AI components into a closed-loop decision support system.
- Explainability and Federated Learning Analysis: This survey uniquely foregrounds the role of Explainable AI (XAI) tools, specifically SHAP and LIME, and Federated Learning in enabling interpretable, privacy-preserving water-quality monitoring, aspects underrepresented in prior reviews.
- Identification of Open Challenges and Future Directions: The review identifies six critical research gaps: geographic generalizability, benchmark inconsistency, black-box opacity, temporal modelling limitations, rare-event data scarcity, and edge deployment. It outlines concrete pathways for future research involving Physics Informed GANs, Large Multimodal Models, and Generative Digital Twins.

2 Literature review

The existing body of literature on AI-based water-quality assessment has grown substantially over the past decade, spanning classical statistical models, ensemble methods, recurrent and convolutional neural networks, and, more recently, transformer-based and generative architectures. This review synthesizes 107 peer-reviewed studies selected through a rigorous PRISMA protocol to present a coherent, evidence-based mapping of algorithmic progress, dataset diversity, evaluation practices, and application domains in water-quality monitoring (Table 1).

2.1 PRISMA flow

Initially, 351 articles were retrieved prior to screening from Springer Nature (67), Elsevier (64), IEEE (56), Taylor and Francis (26), ACM (20), MDPI (19), Wiley (19), IOP Publishing (14), IWA Publishing (11), Hindawi (4), and other sources (51). Chronologically, this initial corpus spanned 2016 to 2026, comprising 6 articles from 2016, 23 from 2019, 7 from 2020, 79 from 2021, 83 from 2022, 61 from 2023, 33 from 2024, 36 from 2025, and 23 from 2026. Keywords like "Water Quality assessment using AI", "Advancements in AI-Based water quality Prediction", "Water potability prediction using Deep learning", "Water Quality Analysis using Machine learning", "IoT based Smart Water Quality analysis", "Water quality assessment using Federated learning", "Machine learning approaches in water contamination prediction", "Heavy metal prediction in water quality using Deep learning", "Algal bloom forecasting using IoT based AI approaches", "Water quality assessment using GAN", "Use of SHAP and LIME in AI assisted water quality monitoring", and "Self-Attention based Water quality analysis" were used. Only articles in English were used in this search and this survey. Later, these were assessed manually for identifying duplicates. A total of 30 records were found to be duplicate and were removed, alongside 4 records that were excluded for other reasons. After deduplication, the remaining 317 articles were screened in two stages: one based on title and abstract and another stage with full-text screening. From this, 73 articles were removed in total, as 38 records did not relate to water quality whereas 24 records did not discuss AI methods and 11 records were non-empirical. Full-text assessment was done on 244 articles and it was found that 48 records were identified to be duplicate, using automation tools like Rayyan. 34 records were technically insufficient, 32 records were not primarily for water quality, and 23 records were low-quality studies, so a total of 137 were excluded as they were not relevant to research objectives or aims. For the final review, 107 articles were included: 1 article from 2016, 3 articles from 2019, 1 article from 2020, 21 articles from 2021, 27 articles from 2022, 21 articles from 2023, 15 articles from 2024, 14 articles from 2025, and 4 articles from 2026. The final review consists of 107 records from publishers Elsevier (38), Springer Nature (35), IEEE (12), IWA publishing (4), Taylor and Francis (4), MDPI (4), Hindawi (2), Wiley (1), ACM (1), IOP publishing (1), and others (5). Figure 4 shows the journals that were selected, excluded, and used for final review (R2:C).

2.2 Data and real time monitoring

Water quality indicators are typically classified into physical, chemical, biological, and radiological categories. These indicators help to identify the quality of water. Figure 4 explains these terms. Implementation of real-time sensor data will help in identifying potent risk in drinking water. Researchers used K-means Clustering to classify water samples into distinct types, namely mud water, lemon water, salt water, drinking water, and municipal water, which were trained on labeled datasets. Two sensors, namely pH and TDS meter, were inserted into water and used hardware components like Arduino with Raspberry Pi 3 processor for prediction and clustering. Cluster assignment correctness and visual analysis of pH vs. TDS plots were used for evaluation. The article confirms clear clustering for each water type class, showing model's capability to distinguish them (Kolli et al., 2023). To reduce human intervention, a study to integrate edge computing, cloud analytics, and sensor data (IoT-based framework) combined with an ML model to classify water quality and automate alerts was carried out. The Random Forest Model was found most suitable as it was the best performer with 88% accuracy. The framework consisted of Raspberry Pi for edge computing and to visualize

TABLE 1 AI model categories applied to water quality assessment.

Ref and year	Indicators and target	Key models/ algorithms	Main finding	Evaluation metrices	Limitations	Reported performance	Dataset location
Najafzadeh et al. (2019)	Input: Ca^{2+}, Na^{+}, Mg^{2+}, NO_2^{-}, NO_3^{-}, PO_4^{3-}, Electric Conductivity, pH, Turbidity. Target: BOD, COD and DO	GEP (Gene-Expression Programming), Model Tree, EPR (Evolutionary Polynomial Regression)	Nonlinear relationships between BOD, COD, DO are established by evolutionary models.	R, RMSE, MAPE, NSE	Models show moderate prediction performance on Dissolved Oxygen.	EPR performed best for COD and DO prediction with R of 0.84 and 0.71 and GEP showed lower errors in BOD with R of 0.86	Karun river, collected at 8 stations for 16 years
Najafzadeh and Niazmardi (2021)	Inputs: Electrical Conductivity, Na^{+}, Ca^{2+}, Mg^{2+}, phosphate, nitrite, nitrate-N, turbidity, pH Target: BOD and COD	Random Forest, Support vector Regression and proposed model MKSVR (Multiple-Kernel Support Vector Regression)	Representations of nonlinear relationships were improved by combing multiple kernels	R, RMSE	The proposed model outperformed only on one target BOD. As conventional SVM performed well in COD prediction	MKSVR showed best result in BOD prediction with R^2 of 0.80 while traditional SVM showed only 0.68.	Karun river, Iran
Mohd Zebaral Hoque et al. (2022)	Input; pH, DO, CO, BOD, NO_{3-}, FC Target: WQI	Decision Tree, Linear and ridge regression, lasso, SVR, Random Forest, ANN	Improved WQI prediction from water-quality rating features	Correlation coefficient, MSE	Highly dependent on derived/ engineered WQI	Best (Linear and ridge) MSE of 0 and Correleation Coefficient of 1	12 derived parameters with 6 original parameters from Indian rivers (Historical dataset)
Choudhary et al. (2025)	Inputs: pH, DO, BOD, conductivity, nitrite/nitrate, total coliform, and fecal coliform Target: WQI	XGBoost, CatBoost, Gradient Boosting Machine, Random Forest, AdaBoost + linear regression meta-learner, SHAP XAI	WQI prediction increased using stacked ensemble methods, SHAP shows high feature influence and interpretability	Adjusted R^2 and R, MAE, RMSE	Only 390 records are used as dataset used consists of sparse microbial measuresment	Stacked ensemble R^2 reach upto 0.99 and adjusted R^2	Indian river samples from kaggle
Slatnia et al. (2026)	Input: pH, Temperature, DO, BOD, TDS and turbidity	SVR vs. 1D-CNN	1D-CNN demonstrates high ability in learning nonlinear relationships	R^2, RMSE, MAPE	WQI is analytically calculated for external validation is required	1D-CNN outperformed with R^2 of 0.9962	Tilesdit Dam, Bouria, Algeria
Nishat et al. (2025)	Input: DO, pH, TDS, EC, CL, Fe Target: WQI	DT, LR, Ridge, SGD, SVR, RF, XGB, ElasticNet, Bayesian Ridge, CatBoost, Extra Tree, LightGBM, and ANN	Comparison and feature selection shows efficiency of capturing Relationships by many ML models	R^2, NSE, MAE, RMSE	A locally tailored WQI framework is unavailable.	ANN outperformed with R^2 of 0.971	4 Dhaka rivers: Balu, Burigana, Tongi from 2001–2023

(Continued)

TABLE 1 (Continued)

Ref and year	Indicators and target	Key models/ algorithms	Main finding	Evaluation metrices	Limitations	Reported performance	Dataset location
Elmotawakkil et al. (2025)	Inputs: Temperature, pH, turbidity, Salinity, DO percentage and concentrations, depth. Target; WQI and WQC	ANN, DT, SVM, XGBoost, and Decision Tree vs. LSTM	Demonstration of effectiveness in combining a Classification and regression AI approaches	Accuracy, Precision, recall, F1, Kappa, MSE, RMSE, R^2	Overfit Identification, limited generalization	LSTM outperformed with R^2 of 0.9999 And XGB with high accuracy 99.83%	Obtained from Github repository
Wang et al. (2024)	Inputs: Meterological variable, pH, DO, TN, TP, EC, Turbidity, KM_nO_4, Ammonia Nitrogen. Target: DO	EEMD-CNN-BiLSTM, GRU, Informer, and LSTM	Multi-step DO forecasting is improved in this hybrid approach as EEMD reduce noise, BiLSTM and CNN captures temporal dependencies and local features	Coefficient of Determination, MSE, MAE, MAPE	Experimentation required for EEMD modal-number selection	EEMD-CNN-BiLSTM outperformed well with 5 to 7% improvement in R^2	Sanhongcun Village monitoring stations, liaohe River, China
Mu et al. (2026)	Multivariate time series data is used to forecast water quality across multi horizon	RNN, LSTM, Informer, PatchTST, PiTNeT (Transformer)	Captures long-range temporal dependencies and reduced error accumulation by PDE-based physical constraints	RMSE, MAE, MAPE	Validation required as physics-informed formulation requires physical constraints	PiTNet outperformed all models with least Errors (15 to 40% reduced)	Zongguan monitoring station, Yangtze River Basin, China
Rathinam (2025)	Inputs: pH, TDS, river discharge, Bicarbonate, chloride, sulfate, sodium, magnesium, calcium	GAN (QGCNN-X), ANN, KNN, RF, SVM, DT, GB, AdaBoost, VAE, WGAN, Naïve Bayes, Diffusion	Computational cost is addressed using Quantization while WQI prediction is improved using hybrid models.	MAPE, Coefficient of Determination.	Computational complexity and needs further validation	QGCNN-X outperformed all other models	Across regions across India
Zounemat-Kermani et al. (2021)	Inputs: Solar radiation, sunhine hours, wind speed, relative humidity, maximum and minimum temperature Target: monthly pan evaporation	Soft computing: MLP-LM, MLP-CG, RBFNN, MARS, M5Tree, SVR Statistical: Kriging, Improved Kriging and Response Surface Method	Improved Kriging from polynomial to exponential form shows high prediction performance	MAE, RMSE, MAPE, R^2, Whitney statistical test, MBE, Willmott index	Research focus on water pan evaporation rather than water quality	MLP-LM showed lowed rates MAE of 0.492 mm and d of 0.981 compared to overall models and every model achieved d > 0.95	Adana and Antakya, Turkey

(Continued)

TABLE 1 (Continued)

Ref and year	Indicators and target	Key models/ algorithms	Main finding	Evaluation metrices	Limitations	Reported performance	Dataset location
Najafzadeh and Basirian (2023)	Input: Turbidty, sodium, sulfate, hardness, potassium, fluoride, DO, chloride, arsenic, alkalinity, pH, nitrate and magnesium Target: WQI	MT, MARS, GEP, EPR,	Calculating WQI by ombination of remote sensing spectral information reducing continuous field measurements	IOA, RMSE, MAE, SI, F-Test, uncertainty analysis, and AUC	Validation need for model transferability, limited ground observation	MARS showed high performance in training with IOA of 0.992 and SI of 0.0073	Hudson river, New York, USA using Landsat-9 OLI-TIRS
Kisi and Parmar (2016)	Inputs: pH, AMM, TKN, TC, FC, water temperature Target; monthly Chemical Oxygen Demand	Least Squares Suport Vector Machine, M5Tree, MARS	Input combination of AMM + TKN + WT improved prediction	Correlation Coefficient, RMSE, and cross-validation-based comparison	Classification of water as potable and non-potable is not explicated, Performance is determined by selected input combinations	MARS slightly perform better than LSSVM with better average RMSE by 1.47 and 19.1% by M5Tree	Nizamuddin monitoring station, Yamuna river, Delhi, India
Kisi et al. (2019)	Inputs: correlation and sensitivity analysis done using groundwater-quality varaibles Target: EC, SAR, TH	Four Combination of ANFIS with CGA, PSO, ACOR, DE	Conventional ANFIS showed substantially improvements with Evolutionary optimization.	R^2, RMSE, MAPE, Sensitivity analysis	Broader geographic validation needed. Computational cost and Complexity is increased by evolutionary optimization	CGA improved ANFIS by Improved EC_R^2 by 0.13, MAPE by 0.59, RMSE by 35.4 and R^2 by 0.14	Isfahan-Borkhar, Iran
Al-Falal et al. (2025)	Inputs: Cr, Cu, Cd, Pb, Fe, Mn, Ni, Zn, DO, COD, NH^4, TDS, pH and temperature Target; HPI, MI, CL and AWQI	MLR, DT, XGB, MLP, AdaBoost, RF, PCA, CA	Group of pollutants and environmental relationships were identified by PCA. (Major contamination contributors were identified to be Cu and Cd)	Wilmott Index, R^2, RMSE, MAE, NSE	Only a limited number of samplings and sites were used, reducing spatial and temporal generalization.	MLP is identified as best model with least errors with WI of 0.98, MAE of 0.05 RMSE of 0.06 and R^2 of 0.99	El Manzala lake, Egypt

and alert cloud portal. IoT Sensors like pH, Temperature, Turbidity, and flow were used (Bhardwaj et al., 2022). Real-time water quality and monitoring are leveraged by LoRa-based communications. LoRa refers to a Long-Range and low-power wide-area network (LPWAN) technology, basically a type of IoT application that enables long-distance transmission at minimal energy consumption. ML models were assessed to identify the best model with feature scaling techniques like Recursive Feature Elimination for finding the optimal features like pH, TDS, EC and ionic constituents for drinking water and sodium-related indices for irrigation water. Of these, Linear Regression achieved highest accuracy for drinking water classification and Support Vector Machines performed best for Irrigation water assessment. This shows that these models can increase the efficiency of smart water quality management systems (Ajayi et al., 2022). A study demonstrated remote sensing spectral information from Lansat-8 OLI-TIRS integrated with ML technique to remove the need for conventional sampling to yield continuous Water Quality assessment (Najafzadeh and Basirian, 2023). A similar study, in which multiple models like M5 model tree, MARS, EPR and GEP, evaluated data obtained from Landsat-7 ETM in which FS-M5MT outperformed other models by classifying Karun river as "relatively bad" in Water Quality classification (Reliability assessment of WQI based on guidelines of national sanitation foundation in natural streams: integration of remote sensing and data-driven models). Further studies combined ensemble models with satellite imagery to provide costal hypoxia (Basirian et al., 2026). Water-quality pollutants based on their type is mentioned in Figure 5. Table 2 discusses the papers reviewed in this article.

2.3 Dataset preprocessing

Data preprocessing is critical, as any dataset inconsistencies can degrade prediction quality and increase error rates. Data normalization, imputation of null values, and the use of PCA (Principal

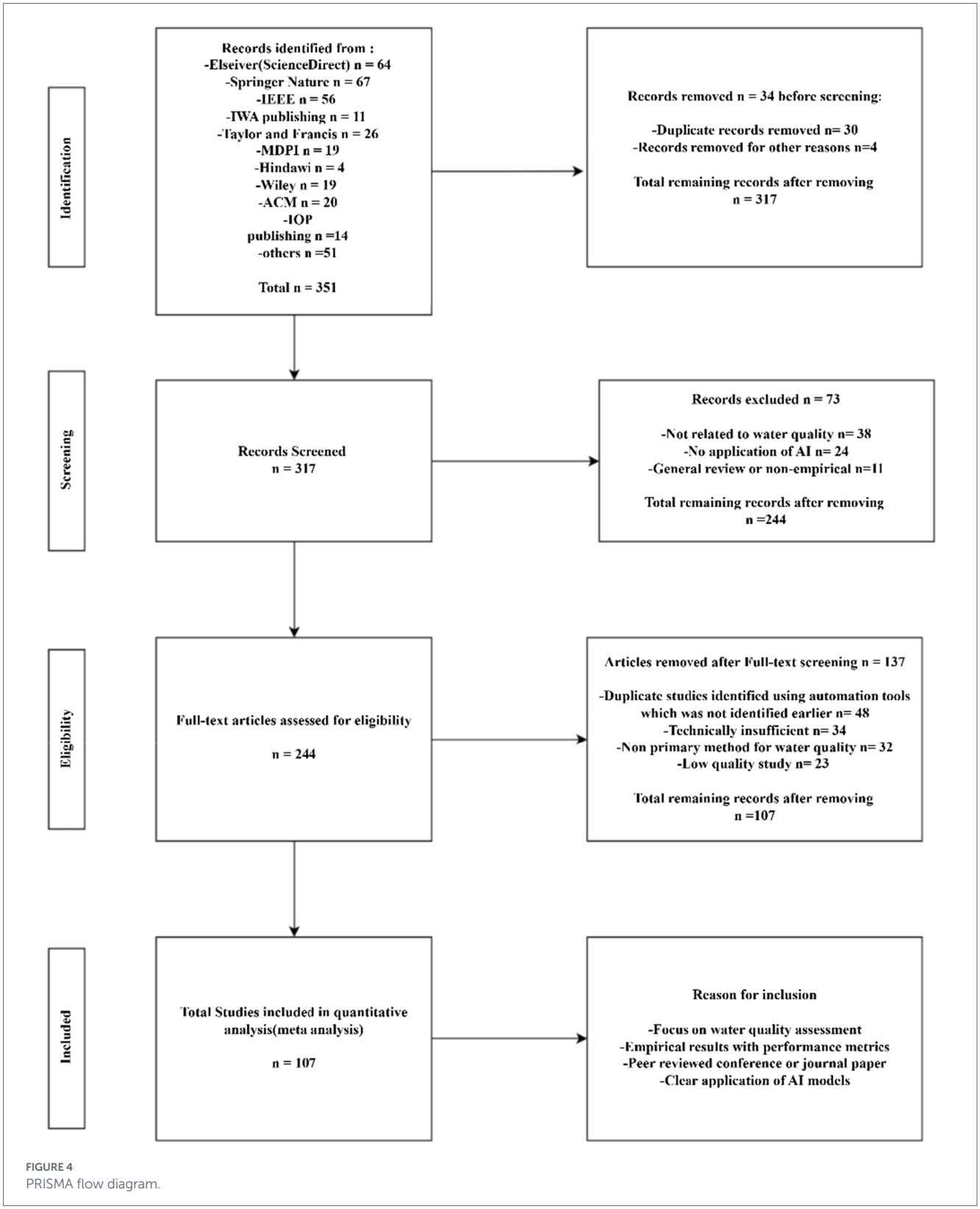


FIGURE 4
PRISMA flow diagram.

Component Analysis) for feature selection result in better outcomes than existing models (Gandhimathi et al., 2023; Khan et al., 2022). Integrating preprocessing techniques such as PCA and Pearson correlation helps select non-redundant features that are highly correlated with the target variables, thereby reducing overfitting (Abuzir and Abuzir, 2022). Predictive performance is optimized using 10-fold cross validation to tune hyperparameters and non-parametric bootstrapping with 199 iterations, which quantified prediction uncertainty (Erickson et al., 2021). Recursive Feature Elimination (RFE) is used to analyze the high informative features. RFE with Linear Regression (RFE-L) and Standardized regression coefficients are used to identify the influence of each parameter on WQI (Sakaa et al., 2022).

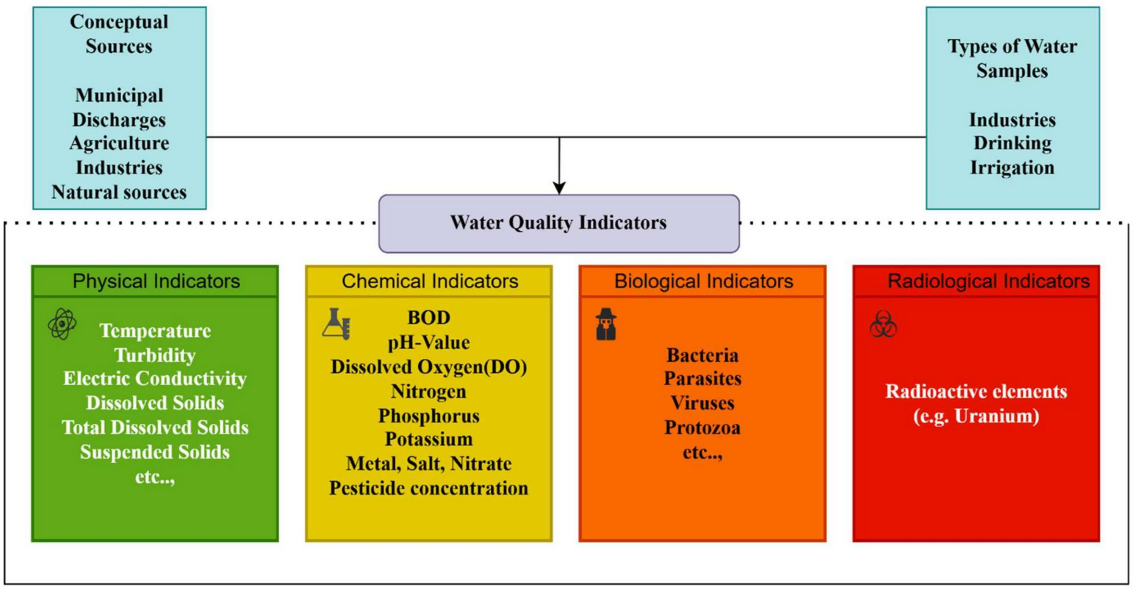


FIGURE 5
Water-quality pollutants.

2.4 Temporal validation and data leakage

Most of the reviewed studies rely on random train and test splits or kfold CV, which risks spatial and temporal leakage in WQ prediction tasks as these data are location-specific and seasonally dependent. Rare usage of rolling or chronological validation resulted in improperly accounted-for seasonal trends. The appearance of same site in both training and testing sets due to the absence of spatial holdout strategies. Information Leakage into the training process from test set by preprocessing methods was applied frequently before splitting. Reported accuracy was inflated as hyperparameter tuning was performed without nested cross validation.

2.5 Types of models

Artificial intelligence models are broadly categorized into machine learning, deep learning, hybrid, and federated learning architectures, as illustrated in Figure 6.

3 Regression-based machine learning models for water quality predictions

3.1 Linear regression

One study compared and applied four regression models, namely Linear Regression, LASSO Regression, Bayesian Regression, and Stochastic Gradient Descent (SGD), and used three physicochemical parameters (pH, DO, and BOD) collected annually from 2012 to 2019 from seven stations to predict water quality. This study demonstrates Linear Regression as the best model in terms of pH parameter as LR thrives on structured and small datasets. SGD performed the best when analyzed using other two parameters (DO, BOD) (Devagopal et al., 2022). One study used a simplified forecasting time series model with Basic Linear, D Linear, N Linear as three variants and compared RNN-based LSTM and GRU models on a pH and EC data of 22,608 samples. CEEMD-LSTF-Linear constantly achieved more accurate results across every metric, as it converts Complex time series data into the simplest form, which is easy for prepossessing and exposes clear linear seasonal trend signals. This enables the model to learn efficiently and outperform deeper models which were prone to overfitting risks (Shin et al., 2025).

3.2 Multiple linear regression

One study evaluated water quality using multiple linear regression (MLR) and multilayer perception (MLP) models. Various other deep learning models were also used, such as Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU). The data were preprocessed using Pauta criterion and min-max normalization. 80% of the data was used to effectively train the model and 20% was subjected to testing where multi source indicators have linear dependencies where LR gains advantage over irregular sewer datasets (Jiang et al., 2021). Further studies were done using Multiple Linear Regression in which the author compared MLR, Gradient Boosting, and Lasso Regression for predicting the WQI and used Classification Models such as Artificial Neural Network (ANN), Support Vector Machine (SVM), and Decision Tree (DT). Multiple Linear Regression was found to be the best in terms of accuracy and ANN showed high performance in terms of water-quality classification. Nine water-quality parameters were taken into account but only four of them were used by the final model (Azrour et al., 2024). One study used eight ML models, namely MLR, ANN, SVM, M5P (M5 Prime- Model tree), RF, LWLR (Locally Weighted Linear Regression), RSS (Random Subspace), and AR (Addictive Regression). The researcher evaluated their dataset of 114 samples across 57 wells between 1999 and 2020 and was evaluated using 12 physicochemical parameters in two scenarios 1 and 2. In scenario 1, MLR outperforms all other models using all features as MLR is robust to irregular and small datasets and achieved nearly perfect correlation of 1. RF performed the best in scenario 2 as it uses only TDS and TH as

TABLE 2 Water-quality dataset.

Ref	Indicators	Place	Missing data treatment	Validation strategy	Data leakage risk	Hyperparameter optimization	Uncertainty Quantification	Reproducibility Notes	Source and size
Tejaswi et al. (2022)	DO, pH, Nitrate, Conductivity, BOD, Fecal Coliform, Total Coliform	No Exact location of sampling is specified, likely from Guntur, Andhra Pradesh,	z-score normalization	70/30 Train/Test	Moderate	MSE-based weight tuning	Histogram error plots, regression R values	Moderate reporter from ANN/LSTM parameters	Real time data, not specified
Deng et al. (2021)	Chlorophyll-a, Total Inorganic Nitrogen, Phosphorus, DO, Water Temperature, Secchi Disc Depth, BOD, pH	Tolo Harbour, Hong Kong SAR	Linear interpolation for missing biweekly/monthly data	Train/validation/ test split	Low (due to clear separation of train and test split)	Gradient descent, LM, GA, and PSO hybrid optimization	Sensitivity analysis of variables	MATLAB implementation described high reproducibility	Environmental Protection Department (EPD), Hong Kong 11,293 daily samples
Feng et al. (2024)	Electrical Conductivity, Turbidity, Water Water Temperature, Target: DO	Environmental Protection Department (EPD), Hong Kong	Normalization, Abnormal data removed	Train/Test split	Low (due to short-term dataset and robust preprocessing)	Grey Wolf Optimizer for SVR parameters	Reported MSE, MAE, R^2	Strong reproducibility detailed from ensemble workflow	IoT sensors deployed in Demonstration Center for Marine Technologies, Shandong Province, China 1,324 samples (10 days, 10-min intervals)
Patel et al. (2022)	pH, Hardness, Solids, Sulfate, Chloramines, Conductivity, Organic Carbon, Trihalomethanes, Turbidity	Indian water quality based sample	Mean Imputations for missing values	Train/Test split 70/30	Moderate as synthetic oversampling may inflate performance	Hyperparameter tuning	Feature importance via LIME	High reproducibility form open dataset	Kaggle Water Quality Dataset 3,276 samples
Mokhtar et al. (2022)	Electrical Conductivity, Sodium concentration, Calcium concentration, Bicarbonate concentration	Bahr El-Baqr Drain, Eastern Nile Delta, Egypt	Standard lab analysis; with no imputation report	75/25 Train/Test split	Low	Grid Search	RMSE and SI reports	Moderate reproducibility from clear methodology	Field sampling from Bahr El-Baqr drain, Egypt 105 water samples

(Continued)

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TABLE 2 (Continued)

Ref	Indicators	Place	Missing data treatment	Validation strategy	Data leakage risk	Hyperparameter optimization	Uncertainty Quantification	Reproducibility Notes	Source and size
Mamat et al. (2021)	DO, BOD, COD, AN, SS, pH	Langat River Basin, Kajang, Selangor, Malaysia	Not specified	k-fold cross validation	Low (CV reduce leakage)	Kernel tuning	NSE, RMSE, R reporter	Strong reproducibility described from algorithm flow	Department of Environment, Malaysia 10 years monthly data
Zhang et al. (2021)	Turbidity	Tai Lake Waterworks and Huzhou Meteorological Observatory	EMD decomposition, Outlier rejection	90/10 Train/Test split	Low (due to robust preprocessing)	Random Forest Parampara tuning	Morris method (sensitivity analysis)	High reproducibility from python implementation	Tai Lake, China 3 years hourly data
Zhang Z. et al. (2022)	Total Nitrogen, Nitrate Nitrogen, Ammonia Nitrogen, Total Phosphorus	Chesapeake Bay Program	Not specified	10-fold cross validation	Low	Random Forest hyperparameter tuning	SHAP analysis	Strong reproducibility from a public dataset	Potomac River Basin, Chesapeake Bay Watershed, USA
Shams et al. (2024)	BOD, DO, pH, Conductivity, Nitrate, Fecal Coliform, Total Coliform	Lakes and rivers across multiple locations from India	Normalization, Mean imputation	Cross validation	Moderate (due to overfitting risk)	Grid search	MAE, R^2, MCC	Reproducibility high due to modest dataset size	Kaggle Water Quality Dataset 1,991 samples
Ahmed et al. (2023)	DO, pH, TDS, Turbidity, Water and Air temperature, Rainfall, Humidity	Keane Bridge monitoring station, Sylhet	Interpolation for missing values	Train/Test/ Validation split	Low	MODWT decomposition + MARS tuning	R, WI, MAE	Strong reproducibility detailed from workflow	Surma River, Sylhet District, Bangladesh 3 years' of daily data

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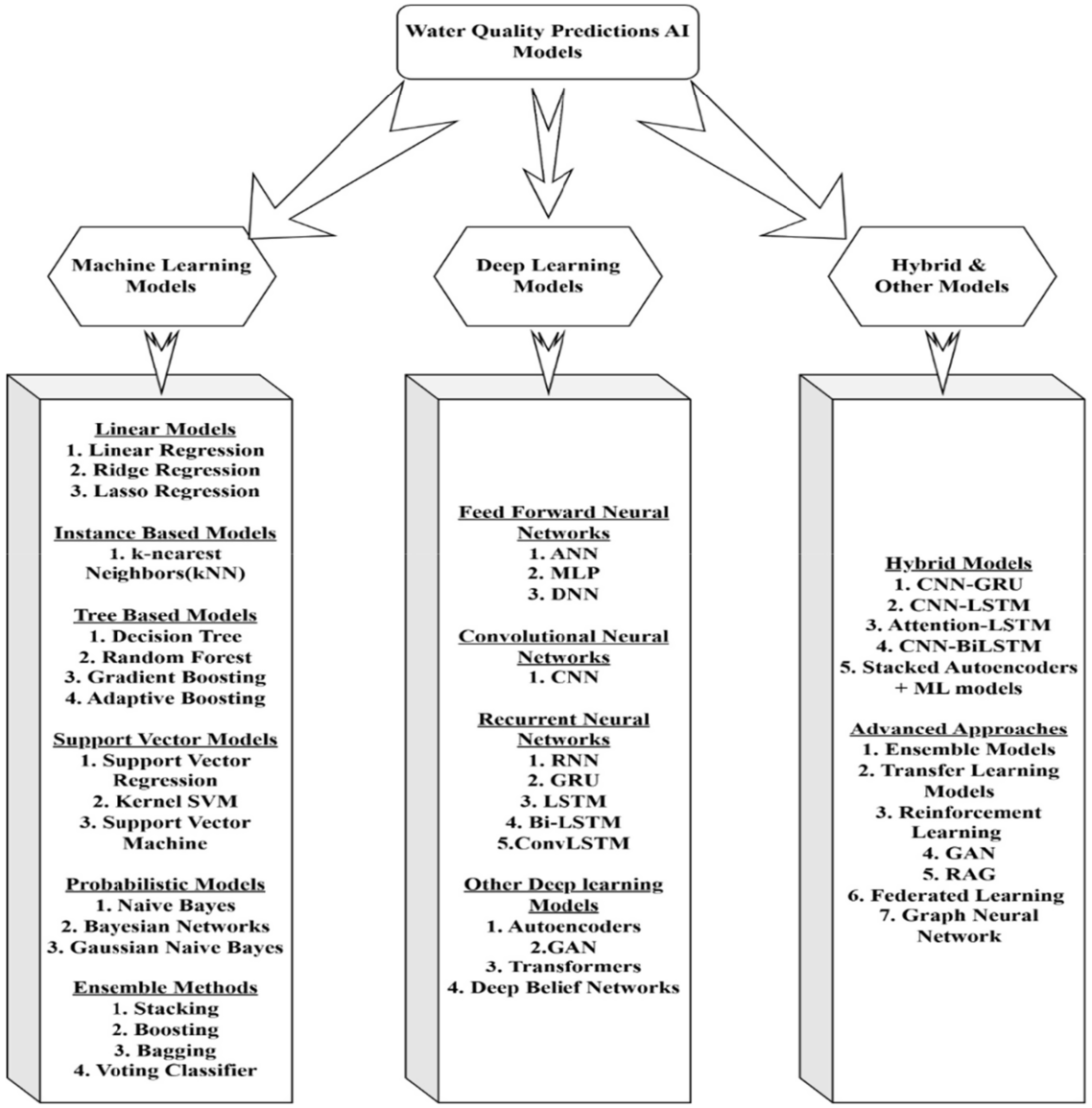


FIGURE 6
AI models used in water-quality prediction.

parameters to evaluate the quality of water (Kouadri et al., 2021). The study also compared ANN (Artificial Neural Network), LSTM (Long Short-Term Memory), and MLR to analyze groundwater quality in scenario 2. Scenario 1 analyzed all variables and identified ANN as the superior model. Scenario 2 analysis on reduced variables showed MLR to surpass other models in accuracy, as smaller datasets used from Akton Basin of 140 samples favored MLR model and irrigation indices like SAR, RSC, PI, KR are algebraic ratios with linear dependencies (Kouadri et al., 2022).

3.3 Other regression models

One study showed that applying Gaussian Process Regression (GPR) achieved 95% in Confidence intervals. This research estimated uncertainty at each stage of the WQI model. Research on coastal water indicators like pH, BOD, and DO were tested using various models like Decision Tree (DT), Support Vector Machines (SVM), Neural Network (NN), and other Ensemble Tree Models. Uncertainty was mostly contributed by sub-indexes function shown by Monte Carlo simulations; 10 fold cross validation shows GPR as the best-fit model with highest accuracy and low errors, as uncertainty propagation was reduced by GPR when compared to other probabilistic models (Uddin et al., 2023; Najafzadeh and Ghaemi, 2019). Additive Regression (AR), in which the effects of individual predictions affects its own results when summed rather than interacting, was found to be more efficient than M5P tree model, Random Subspace (RSS), and Support Vector Machine when tested with a nine-input combination from 35 samples from Akton basin. This dataset was small, linearly correlated, and reduced dimensionality by variable elimination in a fivefold cross-validation approach (Elbeltagi et al., 2022).

3.4 Ensemble models for water-quality prediction

3.4.1 Bagging

In 2006, quantile regression forests (QRF) were introduced to extend standard random forest (RF) models. QRF capture the

heteroskedasticity and non-linearity patterns in regression tasks. Principal Component Analysis is a filter method used for feature selection and can reduce noise and dimensionality on monthly data collected from 2004 to 2013. PCA-QRF exposed latent structures from aligned indicators, outperforming all other models including GA-integrated versions with the least errors for Biochemical Oxygen Demand (BOD) predictions. Since blocking or rolling validation is absent, temporal leakage is likely possible during hyperparameter tuning and feature extraction (Al-Sulttani et al., 2021). Other papers have suggested that RF dominates among most of the supervised learning algorithms on large datasets of 10 years with 5,000 imbalanced records. Multiple models like Random Forest, Decision Tree, Support Vector Machine, and Naive Bayes were compared. Random Forest with an Ensemble of Decision Trees was voted as the best-performing model binary (Drinkable vs. Non-drinkable) with multi-class (Excellent, Very Good, Poor, Very Poor) classification under all metrics as it is robust to noise and handles non-linearity across imbalanced and large datasets better than single decision trees (Prasad et al., 2021). The ensemble of decision trees (Random Forest) with 500 trees, default depth, split parameters, and bootstrap sampling was superior to Multiple Linear Regression (MLR) and an ANN network feed-forward with ReLU activation with Adam Optimizer (for a 20-year study with 252 monthly observations which requires ensemble aggregations on non-linear long term seasonal patterns). Shapley Additive Explanations was used for interpretability. Later, SHAP values confirmed DO (Dissolved Oxygen) and BOD (Biochemical Oxygen Demand) as the most influential features for identifying the WQI (Miller et al., 2023). In a comparative study aiming to find the best model for water quality classification (Potable, non-potable), Random Forest, Decision Tree, Support Vector Machine, Linear Regression, and Gaussian Naive Bayes were compared. XAI techniques were used to explain model decisions and identify key parameters. After testing, RF performed the best and was chosen by XAI (SHAP) for its superior performance on large datasets without overfitting using ensemble classifiers under all metrics (Nallakaruppan et al., 2024). Random forests use the bootstrap technique and are trained by random sampling with replacement from real high-dimensional datasets with 28 indicators and 97 watersheds. These predictions are combined (average) to improve accuracy. The RF model showed capturing thresholds on partial dependencies plots when tested using cross validation. On comparison with another ensemble method, Boosted Regression Tree (BRT) RF slightly outperformed all the other models in all indicators with low errors (Alnahit et al., 2022).

3.4.2 Boosting

Another ensemble learning technique is boosting. Techniques like CatBoost are highly robust in categorial predictions. In a study involving a spatial temporal dataset of 15 beaches with routine monitoring to predict the presence of bacteria, CatBoost performed well in overall predictions including General Model, spatial prediction, and temporal predictions by achieving higher R^2 when compared with other ensemble techniques like XGB and RF. CatBoost captured the nonlinear dynamics of fecal bacteria and reduced prediction bias. The full dataset is used for feature engineering and by SHAP before split temporal leakage is possible (Grbčić et al., 2022). Automatic hyper parameter tuning Boosting techniques like CatBoost are integrated with Stimulated Annealing to form Self-Optimizing CatBoost Algorithms with spatial mapping for ensemble flexibility across mixed polluted urban rivers. This provides various advantages over UAV and satellite data by improving generalization to multisource remote sensing data and avoiding overfitting. Some supporting techniques were also used like Feature enhancements by Pearson correlation, Spatial mapping, and Band filtering. It is noted that without spatial mapping, traditional satellite inversions yielded very low R^2. Satellite data has a risk of spatial leakage with random validation as the same river samples are used for both training and testing (Chen et al., 2023).

3.4.3 Stacking

A stacking ensemble model involves using a meta model (high-level model) to validate the outputs from first stage (level 0) evaluation as inputs for the next stage (level 1) to identify methods and improve overall performance. A three-base learner model with Gradient Boosting Machine (GBM), Generalized Linear Model (GLM), and Neural Network (NN) gives outputs as inputs to meta-model GLM to analyze and achieve high performance (Talukdar et al., 2023). Multiple studies on meta models utilize many outputs for base models like XGB, RF, and DT, which are stacked together to form a meta model. A multi land climate interaction used a dataset from Godavari basin between 2019 and 2021 and favored ensemble integration as overfitting is minimized by multiple base learners and was preprocessed using AutoML. This meta model's robustness was improved by hyperparameter tuning, giving highest R^2 scores in separate parameter-based water quality predictions. Land and climate inputs are prone to temporal leakage as these inputs are not chronologically blocked (Satish et al., 2024).

3.5 Hybrid ML models

Most ensemble voting classifiers determine final predictions through majority voting, selecting the class with the highest number of votes or the highest average predicted probability. This is better than a single model, which can result in overfitting. A combined model of Gradient Boosting, AdaBoost, Decision Tree, and Random Forest with Recursive Feature elimination with Cross-Validation and Particle Swarm Optimization techniques works well in suitability predictions of aquaponic farming as voting classifiers handle imbalances with SMOTE on aquaponic water datasets with multiple indicators (Yadav et al., 2025).

3.6 Evolutionary ML model

Evolutionary models like ERP and GEP are inspired by natural evolutions and are data-driven ML methods. One study showed that GEP performed better in predicting the species richness and EPR was best for predicting River Macrophyte Nutrient Index from 200 samples of a Polish river. But these models could not perform on all five ecological indicators as they are target-dependent and sensitive to hydromosphological conditions, which limits their generalization (Najafzadeh et al., 2024).

4 Deep learning models

Deep Learning Models consist of Neural Networks made up of connected neurons (Perceptron) which are inspired by a human brain, shown in Figure 7.

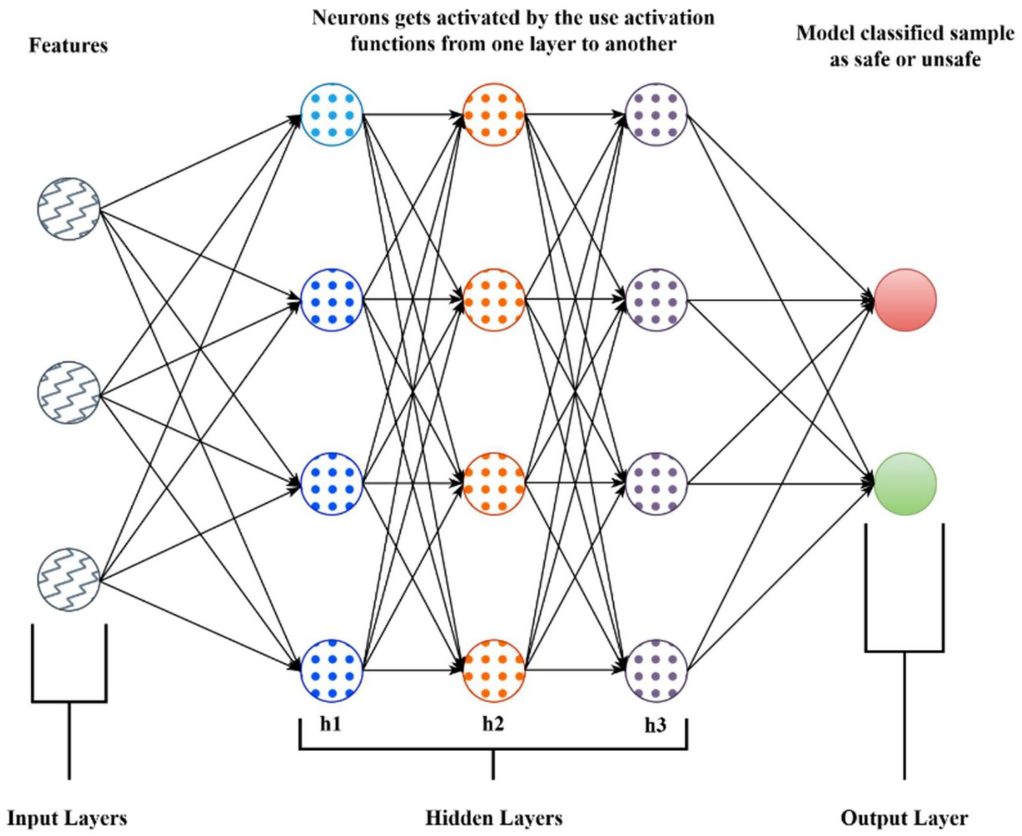


FIGURE 7
Overview of a deep learning model for water-quality prediction.

One study surveyed a wide range of deep learning architectures. Models evaluated a multiple marine water quality task. Each model's performance varied from task to task. LSTM outperformed in Chlorophyll-a proxy for algal biomass, PCA-LSTM (Principal Component Analysis Long Short-Term Memory) performed best for Dissolved Oxygen, CNN + LSTM Models had the highest accuracy for Harmful Algal Blooms, Sea surface Temperature prediction was best performed by GRU-CNN Models, CNN also performed well on Oil Spill detection as they are good at image-based tasks, and Anomaly Detection of Turbidity was best detected by Wavelet ANN. This proves that each model excels at certain tasks (Gambín et al., 2021). One study compared classification models on 1,679 samples across Indian states with regression models to predict the WQI. Classification models were evaluated using accuracy and Regression Coefficient. It is noted that FFNN (Feed-Forward Neural Network) preprocessed using z-score performed the best, achieving 100% accuracy in the classification of water Quality due to its adaptive thresholding and capability to capture nonlinear mappings (William et al., 2023). Deep learning is also advantaged when handling large-scale datasets. A H2O Deep learning model (Multiple layer feedforward neural network with rectifier activation, cross entropy loss, and adaptive learning rate) was found to be overall best performer with two scenarios by generalizing across non-stationary and non-linear datasets. One dataset had a small catchment of five monitoring stations and two with a large catchment of 19 monitoring stations. This DL model consistently performed well with small and large monitoring stations with balanced precision (Tiyasha et al., 2021). One study constructed a fully connected Deep Neural Network trained to predict heavy metal parameters from Taiwan with 7,702 samples powered by virtual sensors with reduced complexity that is more cost effective than relying on a expensive lab equipment. A Relu activation function, which is one of the fastest models and can work well on combined parameters, recorded and predicted WQI across all Heavy Metal Evaluation Index (HEI) and Entropy-weighted Heavy Metal Contamination Index (ECHI) categories. The model used normalization and batch training for preprocessing and stabilized learning (Moeinzadeh et al., 2023). A comparative study of 392 samples from Haryana compared two Ensemble Machine learning models XGBoost with regularization and Gradient Boosting Machines with a Deep Neural Network with Adaptive learning rate, dropout, and rectifier activation to identify the Entropy WQI and WQI, and identified pH has the least and EC the most influential features. DNN consistently outperformed the ensemble models in reliability and accuracy, as DNN models learn hierarchical and non-linear patterns more efficiently due to deeper representation learning than simple tree-based models (Raheja et al., 2022). In another study, Feature Importance was assessed using Pearson Correlation to conduct sensitivity analysis and identified NH3-N and COD as key indicators during lockdown from Putrajaya Lake monitoring station. Multi-Layer Perceptron showed the best results with least difference from predicted to actual Water Quality Index as MLP captures relationships between nonlinear parameters like NH3-N and COD. Analysis was also done to check the changes in WQI by MCO (Malaysia's COVID-19 Movement Control Order), which showed improved Class II to Class I during MCO (Najah et al., 2021). One proposed model consisted of Bi-directional Long Short-Term Memory for time-series forecasting designed for Water Quality

Assessment (DLBL-WQA) of monthly samples over 6 years. An architecture with Feature extraction by Convolutional Neural Network, Mean value method for Missing value imputation and to reduce training error, and Gaussian likelihood-based Optimized loss function was used. DLBL-WQA performed well with the fewest errors by missing value imputation and leveraged learning of bidirectional temporal dependencies more effectively than other comparative models (Khullar and Singh, 2022).

4.1 Convolutional neural networks

Several studies demonstrate that convolutional neural networks (CNNs) are particularly effective for image-based prediction tasks. One study analyzed water quality with visual parameters like Color, Clarity, and Texture using a Primary Convolutional Neural Network to rescale and classify images as safe or unsafe. This model was later validated using separate image data subsets consisting of 200 Google earth images. When compared with other classification models, CNN not only was identified best in terms of performance accuracy but is also cost-effective as it captures spatial patterns in images (Vijay Anand et al., 2023). Certain types of CNN models suffer from performance degradations in longer predictive horizons. A Multi-Input Multi-Output Temporal Convolutional Network (MIMO-TCN) model with Dilated convolutions for lang-range temporal dependencies was examined. These structures are casual and fully convolutional and mitigate vanishing/exploding gradients skip connection. Repeat vector was used to achieve multi-step output generation. Though the runtime decreased significantly, the accuracy degraded along the increasing horizon (Zhang and Li, 2023).

4.2 Artificial neural network

Gated-LNN, which is a combination of gating mechanism and Liquid Neural Network (LNN which are flexible over time), consists of fully connected layers that can be used in both regression (WQI prediction) and classification (Water Quality Classification). They performed better when combining gating with temporal learning and liquid dynamics than traditional models, with high R-value (0.9995) in regression tasks and high F1-score (99.76%) in Classification by using Adam Optimizers, L2 weight decay, and dropout on 1897 Indian water samples (Chadalavada et al., 2025). A Multilayer Perceptron with Sequential Feature Selection (SFS, which initially finds the subset by adding up features one by one) on a large European water base dataset uses a GA hyperparameter tuning to capture nonlinear relationships and optimize hyperparameters in identifying the Biochemical Oxygen Demand better than all other models (Ooi et al., 2022). Studies were conducted to identify the best deep learning models for classification. Many models were implemented and compared. CNN (1D convolution), RNN, BPNN (Multilayer perceptron with backpropagation), LSTM, Bi-LSTM, GRU, and ANN (Standard feedforward neural network for classification) were used to categorize water into five categories: Excellent, Good, Poor, very Poor, and Not Suitable. In all metrics, Precision ANN shows ability to generalize well across diverse ground water datasets of 3,000 samples from Indian rivers and excelled greater accuracy (DSouza et al., 2023). A MLP (Multi-Layer Perceptron) Artificial Neural Network (ANN) model containing three layers with 10 neurons in a hidden layer was trained on El Pedregal water utilities from the Peru water distribution network with the help of the Scaled Conjugate Gradient algorithm. To reduce predicted failure risks in water quality, sensitivity analysis was conducted to asses the influence of each input features. This model captured nonlinear infrastructure and showed robust generalization in prediction (Dawood et al., 2021).

4.3 Recurrent neural network

One study proposed a hybrid framework combining an attention-based gated recurrent unit (GRU) for time-series forecasting with a randomized search gradient-boosted regression tree (RS-GBRT) for spatial prediction and hyperparameter optimization. The attention-GRU performed better in central monitoring point and RS-GBRT in terms of Three-dimensional Prediction of DO with least error values (Cao et al., 2021). Variants like VVR-GRU and VRR-GRU are prone to information leakage, which reduces performance efficiency lower than a single GRU. SO RRR-GRU (Rolling Decomposition + Recursive Reconstruction Gated Recurrent Unit) is proposed, where information leakage is prevented using rolling decomposition to reduce prediction errors and ensure realistic performance. Only essential parameters were analyzed by Spearman correlation. This model can handle high variance and also increases generalization to real unseen data (Yan et al., 2023).

4.4 Long short-term memory

LSTM can use EMD (Empirical Mode Decomposition) to decompose the time series into IMF's (Intrinstic Mode Functions) retaining outliers, making it easy for Models like LSTM to learn complex patterns. EMD-LSTM dominate over other models like TO-LSTM (Traditional Operation + LSTM), STFT-LSTM (Short-Time Fourier Transform + LSTM), EMD-PLSR (Partial Least Squares Regression), EMB-GBR, and EMD-DNN on large mixed frequency data of 35,712 points from drainage networks. Any outlier retention in data has a risk of leakage and so frequency alignment is done before chronological split using rolling block validation. EMD preprocessing stabilized fluctuated and non-aligned data, enabling LSTM to learn patterns to utilize data, showing increased efficiency in complex urban drainage signals (Zhang Y. et al., 2022). LSTM (Long Short-Term Memory) are designed to capture long-term dependencies, which can be enhanced by integrating Adaptive PID (Proportional Integral Derivative) controller to adjust aeration in real time to improve aquaculture health and minimizing operational costs in aquaculture systems. This proposed model adapts to environmental variability better than static threshold models and is shown to have a higher performance in multiple metrics than other models (Alluhaidan et al., 2025). LSTM models can be constructed with Encoders and Decoders (seq2seq models) for long-term dependencies capturing in multi-step forecasting. SFG filters (Savitzky–Golay Filter) are also used to remove noise without changing time series characteristics. A large dataset with 10,000 + data samples ensures robust learning. This SE-LSTM model consistently performed well, especially with multi-step prediction by reduced noise in CODMn and DO indicators (Bi et al., 2021). LSTM is fused with a TCN (Temporal Convolutional Network) to achieve short-term temporal features; this can be parallelized with LSTM, which handles long sequence memory. This combination captures faster and more subtle patterns before LSTM processes them. But using an attention mechanism can degrade LSTM-TCN function as TCN already focuses on its dilations, meaning attention re-weights time steps and overwrites TCN learned patterns. TRMF handled missing values with little risk of data leakage in 27,077 samples, increasing the efficiency of LSTM to show superior performance (Li et al., 2022).

A combined model of Gradient Boosting Decision Tree (GBDT), which identifies the key features influencing prediction (feature selection), with LSTM formed a hybrid model GBDT-LSTM. The LSTM parameter was tuned by Cross-validation grid search like Epochs, batch size, learning rate, timestep, and number of trials for a DO (Dissolved Oxygen) prediction in a fishery water model. During validation, the model outperformed all models like Extreme Learning Machines, Backpropagation Neural Network, Particle Swarm Optimized Least Squares Support Vector Machines, Standalone LSTM, and non-optimized GBDT-LSTM. Also, evaluation results show robust fitting abilities and generalizations (Huan et al., 2020). In a study to predict stream water quality from a monthly dataset of 43 samples from Taihu lake basin in 2017, Self-Organizing Map (SOM) combined with K-Means clustering was used to classify sub-basins spatially. Long Short-Term Memory (LSTM) networks capture temporal data in sequence classification like nutrient concentrations for Total Nitrogen and Total Phosphorus, which are altered by landscape fragmentation and human activity. This was quantified using a geographic detector and spatial validation. This has possible risk of data leakage if SOM clustering and landscape metrics were obtained before LSTM training from full dataset (Wang Y. N. et al., 2023).

4.5 Hybrid models of deep learning

Researchers have developed several hybrid architectures to improve predictive accuracy. For example, Huang et al. (2021) proposed CEEMDAN-LZC-GOBLPSO-GRU, a framework that couples complete ensemble empirical mode decomposition with adaptive noise and Lempel–Ziv complexity (CEEMDAN-LZC) for feature extraction, a gated recurrent unit (GRU) for time-series forecasting, and generalized opposition-based learning particle swarm optimization (GOBLPSO) for hyperparameter tuning. Tested on a dataset of 8,784 samples from 2019, this model significantly outperformed baseline CEEMDAN-LZC variants of GRU, LSTM, and RNN. Other Three-Stage Hybrid models like MF-RNNs-EWT-BEGOE were also found to be consistently outperforming baseline and state of the art models like SVM and DBN, which have MF feature selection done by Multi-factorial analysis using the Grey Relational degree method, Replicator Neural Network for outlier detection, Empirical Wavelet Transform (EWT) for adaptive decomposition of time-series data, and BEGOE for an ensemble model combining five predictors like Broyden-Fletcher-Goldfarb-Shanno, Elman Neural Network, General Regression Neural Network, Outlier-Robust Extreme Learning Machine, and Extreme Learning Machine. Robust forecasting on DO is achieved by multi model ensemble optimization and adaptive decomposition forms (Liu et al., 2021). In a related study, Jamshidzadeh et al. (2024) developed a hybrid CNNE-LSTM-GPR architecture, in which a convolutional neural network with enhanced layers (CNNE) extracts spatial features, an LSTM network models temporal dependencies, and Gaussian process regression (GPR) provides interval predictions and uncertainty quantification. Parameterized using rat swarm optimization, the model achieved superior performance across both point and interval prediction metrics when forecasting electrical conductivity (EC) and total dissolved solids (TDS) in a coastal aquifer. It is noted that a CNN model is best for processing visual data like water color, level, and turbidity from images and a LSTM model is better at capturing temporal patterns in sensor data like temperature or pH values. Both are combined together and a hybrid model is proposed which performed best across all datasets with 80% of training and 20% of testing (Jiang and Yan, 2022). CNN, which extracts local features from multivariate time-series data, is integrated with GRU to capture long-term temporal dependencies along with Attention Mechanism to improve focus on critical features forming a hybrid deep learning model PC-CGA to work on multivariate inputs from real RAS sensor data. After training and validation, this model outperforms comparative models in all parameter settings and metrics. Close alignment between predicted and actual values were noticed in visualizations. This is achieved by selective attention and combination of sequential learning with local feature extraction (Yang et al., 2023). Standalone LSTM often struggles with high-noise tasks. So researchers combined Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEDMAN) with LSTM models to Decompose noisy times-series data to smoother components which is predicted as final predictions by linearly combined outputs. Data were collected from the Xili Reservoir between 2019 and 2021. Adam optimizer was incorporated with 250 Neurons. This Hybrid Model was found to have 50% fewer errors in MAE and RMSE across most indicators (Zhang L. et al., 2022).

4.6 Activation functions

Activation functions are important mathematical functions which decide whether a neural network activates or not. Impacts on three different activation functions namely Tanh, Maxout, ReLU or Rectifier were analyzed in the Artificial Neural Network for a Water Quality Prediction Model. In this analysis, Maxout Activation showed the best result as it outputs the maximum of multiple linear functions followed by Tanh in second, but ReLU, which is widely known for its use for faster training, showed up with high generalization errors (Vijay and Kamaraj, 2021).

4.7 Transformers

Transformers are highly effective in learning problems efficiently and provide possible solutions through their Self-Attention- based mechanism. A new transformer was introduced for multivariate time series prediction. Feature selection was made by tokenization of time series into patches. ASEM (Adaptive Spectral Enhancement Module) is used to filter noise at the domain level. Multi-source attentions were used in the weighting of variables. Intra-token and inter-token dependencies used parallel attention mechanisms (Liu et al., 2025).

5 Federated learning

A distributed machine learning approach is one in which multiple clients collaboratively train a server model without exchanging raw data. Each client trains a local model with private data, with model updates acting like weights that are only exchanged with the central server. These updates are aggregated (Fedavg) by the server to form a global model. These are then implemented using Tensor with a Three-layer Convolutional Neural Network architecture. Running time involved 10 communication rounds, with 30 epochs for each in training. On evaluation, FedAvg performed better than all local models (Vellingiri et al., 2023). One study proposed a ring topology-based Deep federated learning model which provided various advantages like centralized server's bottlenecks elimination and higher scalability

with enhanced communication and data privacy (Kalaivanan et al., 2026).

6 Large language models

Evolutions from traditional machine learning and deep learning models have moved towards data-centric strategies or generative AI. In a study to highlight the contamination of undiagnosed private wells in semi-urban and urban areas with biohazards and heavy metal traces, multiple RAG models were compared with query rewriting, fine-tuned embeddings, and adaptive reranking-using domain-specific datasets from Quebec. Documents were preprocessed with fine-tuning embeddings alongside synthetic QA pairs. Statistical significance testing, expert review, and DeepEval metrics were used for evaluation. Llama 3.1 model's results outperformed other models (El Moussaoui et al., 2025). One study utilized five LLMs, namely ChatGPT-4o, Claude 4, Llama 4, Grok 3, and Gemini 2.5, and removed key parameters such as pH, data/time, and water temperature to evaluate variable importance and identify the presence of Coliform in Irrigation water using sanitizer residuals and Physicochemical variables. While every LLM performed well under different conditions, Gemini 2.5 performed well in the absence of water temperature parameters and identified water temperature as a crucial indicator while others showed confounding effects. This proved LLMs can be used as decision support tools and for key feature identification analysis and cross-model evaluation to ensure reliability in water quality monitoring and other environmental applications (Kiyani et al., 2025). A study on integrating computer vision and LLM utilized a multi-modal approach for On-time water pollution management. This model combined a Raspberry Pi camera setup with YOLOv5 object detection for identifying dead fish, synthetic foams, oil spills, industrial waste, wooden logs, trash, and algal blooms. Visuals with textual explanations along with autonomous alerts and solutions for real-time monitored predictions are achieved by integrating LLM via prompt engineering (Samuel et al., 2024). Satellite and UAV-based sensing are cost efficient and fast when compared with traditional monitoring methods and when combined with machine learning models and XAI methods to get high reliable results on real time and large-scale rural and resource-constrained environments via edge and cloud computing. Stakeholders can get accessible narratives on complex outputs using LLM approach, ensuring trust and transparent scalability (Al Noman et al., 2024).

7 XAI techniques

The integrity and robustness of AI and its applications have frequently been questioned. As they were mostly black-box models, questions were raised on how they came to their conclusions. Thus, researchers came up with XAI Techniques. SHAP (Shapley Addictive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) are the most commonly used XAI techniques. They are post-hoc, model-agnostic attribution methods which are used to quantify an instance using a local linear approximation and the influence of each input variable on the prediction of a given model for a specific instance. The non-stationary nature of water quality systems due to floods, land-use change, and seasonal cycles should be tested for stable feature importance rankings even when a model is restrained. Casual inference techniques are used to cross-validate feature-attribution findings to distinguish plausible causal pathways form predictive associations. These have multiple applications like Neural Network Visualization and Saliency Maps, Transparent Logic-Based Models, Counterfactual Analysis, and Partial Dependence plots. They provide various benefits to water-quality management systems like Contamination alerts, Energy Optimization, Leak Detections, Sensor failure Detections, Anomaly Detection, and much more (Infant et al., 2025). Techniques like SHAP also identify features attribution, thus interpretability of the model is sustained. In a XGB model to estimate nitrate for groundwater, SHAP revealed that the soil, climate, and well depth for common soil, fertilizer, and manure used on specific regions as highly important features in nitrate prediction (Ransom et al., 2022). In a model predicting nitrogen and phosphorous concentrations, Random Forest was found to be superior to other comparative models in identifying the Total Nitrogen (TN) and Total Phosphorus (TP). SHAP was used for interpretability analysis and revealed POP (Population Density), Solar radiation, Digital Elevation Model (DEM), and Water area as key features in predicting Nitrogen and Digital Elevation Model (DEM), POP, and Industrial area Density (DIA) for Phosphorus (Xue et al., 2024).

8 Evaluation metrices

Multiple evaluation metrics are used to measure the accuracy and robustness of AI models. The Generalized Linear Model (GLM), which predicts Water Quality Index by learning the coefficients and estimating the maximum likelihood and contribution of each parameter, is also quantified while controlling the distributional assumptions and variance of an WQI task. This model performed the best out of DNN, RF, and GBM models with low error in all metrics like Root Mean Squared Error (RMSE), Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Logarithmic Error (RMSLE), Mean Relative Deviation (MRD), Coefficient of Determination (R^2), Taylor diagram, and learning curves with 5-fold Cross Validation (Rahaman et al., 2024). Table 3 illustrates common evaluation metrics used in water quality prediction with their purposes.

9 AI architecture for water-quality assessments

The reviewed studies demonstrate that effective water quality assessment requires careful selection of an appropriate AI architecture. Water samples from lakes, reservoirs, and groundwater aquifers, which are slow-varying and static systems, are often well predicted by simpler models, as physicochemical process and spatial gradients evolve gradually and can be captured with minimal temporal lags. Conversely, Lotic systems like streams, tidal estuaries, and rivers pose flashy hydrographs, tidal reversal, amd seasonal monsoonal pulses which demand non-stationarity temporal learning dependencies architectures like Deep learning models and GRU. Attention-based architectures, which are capable of disentangling trend and residual, seasonal components, work efficiently on multi-scale periodicity and seasonality in these hydrological regimes. Land-use heterogeneity and

TABLE 3 Summary of evaluation metrics in water-quality prediction.

Metrics	Valuation	Formula	Usage
Precision	Errors < Higher Precision	$\frac{TP}{TP+FP}$	Validation of how many actual predicted samples are truly polluted
Accuracy	Higher = Better Performance	$\frac{TP+TN}{(TP+TN+FP+FN)}$	Evaluating overall correctness of categorical models (Classification)
Recall	Higher = reduced unidentified contamination cases	$\frac{TP}{TP+FN}$	Identification of True Contamination in water samples
F1-Score	Higher F1-Score = balanced performance	$2\times\frac{Precision\times Recall}{Precision+Recall}$	For imbalanced datasets Recall and Precision are balanced.
R^2	Close to 1 = best fit	$1-\frac{\sum_{i=1}^{n}(yi-\hat{y}i)^2}{\sum_{i=1}^{n}(yi-\overline{y}i)^2}$	Recognize how variability in water quality are explained using models
MSE (Mean Squared Error)	Lower = Better	$\frac{1}{n}\sum_{i=1}^{n}(yi-\hat{y}i)^2$	Squared Prediction Errors are evaluated for spontaneous water quality parameters
RMSE (Root Mean Squared Error)	Lower = Accurate predictions	$\sqrt{\frac{1}{n}\sum_{i=1}^{n}(yi-\hat{y}i)^2}$	Magnitude of prediction errors are evaluated as target variable in similar unit.
MAE (Mean Absolute Error)	Lower = Better	$\frac{1}{n}\sum_{i=1}^{n}\lvert yi-\hat{y}i\rvert$	Y_ i - \ hat{Y} _ i
ROC-AUC Score	Higher AUC = high classification results	Space under ROC curve	Ability of model to classify water samples as safe and unsafe
Confusion Matrix	Misclassification patterns are spotted	Visual representation of predictions	Classification performance are evaluated and analyzed

upstream-downstream connectivity are captured using spatially aware architectures. Rather than treating each monitoring station individually, spatial correlation structures can be encoded by GNN. Remote-sensing monitoring networks pose data gaps and sensor uncertainty, which can be addressed using ensemble models to quantify predictions which are robust to missingness and noise. Deep learning architectures support high sampling frequency, which can be sub hourly to daily. Shallow ANN and Random Forest models favor real-time monitoring systems.

9.1 Quality assessment of the reviewed literature

This assessment consists of eight questions, each scored up to 2 points, yielding a maximum total score of 16 (8 × 2). The scores are rated as inadequate (0), partially reported, (1) or clearly reported (2) for each question. Classification of quality is based on the total scores: High (16–13), Moderate (12–8), or Low (7–0). Research articles are classified and represented in Table 4.

The eight questions used to assess the quality of articles:

- Q1: Description of the Dataset and source.
- Q2: Description of the Data preprocessing methods.
- Q3: Description of the Feature Engineering.
- Q4: Description of the Statistical split and procedure.
- Q5: Description of the Validation.
- Q6: Description of the hyperparameter configuration.
- Q7: Description of the Evaluation metrics.
- Q8: Addressing Generalization, Reproducibility, Limitation, and Uncertainty.

10 Results and discussion

A systematic analysis of the selected literature reveals a clear progression from conventional regression models to highly optimized hybrid networks. Unsupervised edge algorithms are effective for real-time applications and can effectively classify immediate hazards, ensemble methods handle class imbalances and missing parameters effectively, simpler and baseline models are useful for smaller and localized datasets, deep learning frameworks favor a comprehensive set of data, and temporal fluctuations in indicators are captured better by specialized sequential architectures.

In recent years, these techniques have been merged in multi-layered structures to address the shortcomings of individual algorithms. Spatial–temporal networks combine convolutional layers and recurrent blocks to model the local geographic changes and temporal evolution. The so-called "black box" aspect of these deep architectures, however, has come to be addressed through the use of Explainable AI (XAI) tools such as SHAP and LIME. These frameworks validate performance by clearly describing the features that influence model predictions, such as biochemical loads and the surrounding land-use. This transparency in structure has been able to effectively connect the high fidelity of computation with the application to environmental policy.

11 Future research directions

Conventional and ensemble models are reviewed across diverse water bodies. Promising results on super resolution and spatiotemporal gap-filling by diffusion models on environmental raster data like

TABLE 4 Quality assessment and classification on literature based on score.

Ref No	Q1 Dataset	Q2 Preprocessing	Q3 Feature Selection	Q4 Split	Q5 Validation	Q6 Hyperparameter optimization	Q7 Evaluation Metrices	Q8 Generalizations	SCORE	Quality Rank
Devagopal et al. (2022)	2	1	1	2	1	1	2	1	11	Moderate
Shin et al. (2025)	2	2	2	2	2	2	2	2	16	High
Jiang et al. (2021)	2	2	2	2	1	2	2	2	15	High
Kouadri et al. (2021)	2	2	2	2	2	2	2	2	16	High
Kouadri et al. (2022)	2	1	2	2	1	2	2	1	13	High
Uddin et al. (2023)	2	2	2	2	2	2	2	2	16	High
Elbeltagi et al. (2022)	2	2	2	2	2	2	2	2	16	High
Al-Sulttani et al. (2021)	2	2	2	2	2	2	2	2	16	High
Prasad et al. (2021)	2	2	1	2	0	1	2	1	11	Moderate
Miller et al. (2023)	2	2	2	2	2	2	2	2	16	High
Nallakaruppan et al. (2024)	2	2	2	2	1	2	2	2	15	Moderate
Alnahit et al. (2022)	2	2	2	2	2	2	2	2	16	High
Grbčić et al. (2022)	2	2	2	2	1	2	2	2	15	High
Chen et al. (2023)	2	1	2	2	1	2	2	1	13	High
Talukdar et al. (2023)	2	2	2	2	2	2	2	2	16	High
Satish et al. (2024)	2	2	2	2	2	2	2	2	16	High

(Continued)

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TABLE 4 (Continued)

Ref No	Q1 Dataset	Q2 Preprocessing	Q3 Feature Selection	Q4 Split	Q5 Validation	Q6 Hyperparameter optimization	Q7 Evaluation Metrices	Q8 Generalizations	SCORE	Quality Rank
Yadav et al. (2025)	2	2	2	2	1	2	2	1	14	High
Gambín et al. (2021)	2	2	2	2	1	2	2	2	15	High
William et al. (2023)	2	2	1	2	1	2	2	1	13	High
Tiyasha et al. (2021)	2	2	2	2	2	2	2	2	16	High
Moeinzadeh et al. (2023)	2	1	2	2	1	2	2	1	13	High
Raheja et al. (2022)	2	2	2	2	2	2	2	2	16	High
Najah et al. (2021)	2	2	1	2	1	2	2	1	13	High
Khullar and Singh et al. (2022)	2	2	1	2	1	2	2	1	13	High
Vijay Anand et al. (2023)	1	2	1	2	0	1	1	0	8	Moderate
Zhang and Li (2023)	1	2	1	2	0	1	1	0	8	Moderate
Chadalavada et al. (2025)	2	2	2	2	1	2	2	2	15	High
Ooi et al. (2022)	2	2	2	2	2	2	2	2	16	High
DSouza et al. (2023)	1	1	1	2	0	1	2	1	9	Moderate
Dawood et al. (2021)	2	2	2	2	1	2	2	2	15	High
Cao et al. (2021)	2	1	2	2	1	2	2	1	13	High
Yan et al. (2023)	2	2	2	2	2	2	2	2	16	High

(Continued)

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TABLE 4 (Continued)

Ref No	Q1 Dataset	Q2 Preprocessing	Q3 Feature Selection	Q4 Split	Q5 Validation	Q6 Hyperparameter optimization	Q7 Evaluation Metrices	Q8 Generalizations	SCORE	Quality Rank
Zhang Y. et al. (2022)	2	2	2	2	1	2	2	2	15	High
Alluhaidan et al. (2025)	2	1	1	1	0	1	2	2	10	Moderate
Bi et al. (2021)	2	2	1	2	1	2	2	1	13	High
Li et al. (2022)	2	1	1	2	0	2	2	1	11	Moderate
Huan et al. (2020)	2	1	2	2	2	2	2	1	14	High
Wang Y. N. et al. (2023)	2	1	2	2	0	1	2	2	12	Moderate
Huang et al. (2021)	2	2	2	1	0	2	2	2	13	High
Liu et al. (2021)	2	2	2	2	2	2	2	2	16	High
Jamshidzadeh et al. (2024)	2	2	2	2	0	2	2	2	14	High
Jiang and Yan (2022)	2	2	2	1	0	2	2	1	12	Moderate
Yang et al. (2023)	2	1	2	2	0	2	2	1	12	Moderate
Zhang L. et al. (2022)	2	2	2	2	0	2	2	1	13	High
Vijay and Kamaraj et al. (2021)	2	2	2	2	0	1	2	1	12	Moderate
Liu et al. (2025)	2	2	2	2	1	2	2	2	15	High
Vellingiri et al. (2023)	2	1	1	2	0	2	2	1	11	Moderate
Kiyani et al. (2025)	2	0	1	0	0	1	2	1	7	Low
Samuel et al. (2024)	2	2	1	2	0	2	2	2	13	High

10.3389/frwa.2026.1946412

remote-sensing imagery and precipitation came from a minimal number of current studies. But validation across real-world data and comparison with simple augmentation models are limited. Large language Models are used for tasks such as natural language querying of databases or metadata extraction from monitoring reports. There has been no empirically established use of LLM for quantitative prediction of water quality parameters and is considered as exploratory. Digital twins are peer reviewed in a small number of studies but lack rigorous validation comparison between conventional ML/ process-based models. These models require substantial adaptation as their application to time series prediction of water quality parameters remains conceptual. So, the development of standardized reporting guidelines and the validation of generative augmentation for irregular or sparse water-quality datasets are still needed. Explanation stability is used to guide management decisions and establish reliability benchmarks before conclusions can be drawn across multiple models.

12 Conclusion

This survey provides a comprehensive appraisal of recent research on AI-based water quality assessment. This literature covers a broad spectrum of traditional, standalone, hybrid, and statistical approaches like Machine Learning, Deep learning, Transformers, and Federated learning. Missing values and edge computed IoT device-based predictions are well handled by ensemble machine learning models whereas complex temporal patterns are captured and highly processed by deep learning methods.

The evolution in this research area has focused on multi-layered compound hybrid architectures augments through combining uncertainty-aware regression layers with spatial–temporal features. Moreover, the use of Explainable Artificial Intelligence tools like LIME and SHAP offer transparency and identify feature influence. In conclusion, hybrid methods have achieved promising robustness and have shown their ability to handle data volume and scale water resource management in specific applications.

Author contributions

KY: Software, Formal analysis, Investigation, Writing – review & editing, Writing – original draft, Conceptualization, Methodology, Visualization. PK: Supervision, Project administration, Writing – review & editing, Writing – original draft, Formal analysis, Software, Investigation, Validation, Resources, Visualization, Methodology, Conceptualization.

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Conflict of interest

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Generative AI statement

The author(s) declared that Generative AI was used in the creation of this manuscript. In preparing this article, I utilized generative AI and AI-assisted tools including to improve readability and language quality. These technologies were applied with human oversight and control, and I carefully reviewed and edited all AI-generated content. I take full responsibility and accountability for the contents of this work.

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References

Abuzir, S. Y., and Abuzir, Y. S. (2022). Machine learning for water quality classification. *Water Qual. Res. J.* 57, 152–164. doi: 10.2166/wqrj.2022.004

Ahmed, A. A. M., Jui, S. J. J., Chowdhury, M. A. I., Ahmed, O., and Sutradha, A. (2023). The development of dissolved oxygen forecast model using hybrid machine learning algorithm with hydro-meteorological variables. *Environ. Sci. Pollut. Res.* 30, 7851–7873. doi: 10.1007/s11356-022-22601-z

Ajayi, O. O., Bagula, A. B., Maluleke, H. C., Gaffoor, Z., Jovanovic, N., and Pietersen, K. C. (2022). Waternet: a network for monitoring and assessing water quality for drinking and irrigation purposes. *IEEE Access* 10, 48318–48337. doi: 10.1109/ACCESS.2022.3172274

Al Noman, A., Tarafder, M. T. R., Rimon, S. T. H., Ahamed, A., Ahmed, S., and Al Sakib, A. (2024). "Discoverable hidden patterns in water quality through AI, LLMs, and transparent remote sensing," in *2024 17th International Conference on Security of Information and Networks (SIN)*, (IEEE), 1–6.

Al-Falal, A. N. A., Elsayed, S., El Fadaly, E. A., Gaagai, A., Aouissi, H. A., Abd El-baki, M. S., et al. (2025). Aquatic system assessment of potentially toxic elements in El Manzala Lake, Egypt: a statistical and machine learning approach. *Results Eng.* 26:105027. doi: 10.1016/j.rineng.2025.105027

Alluhaidan, A. S., Prabu, P., Aziz, R., and Basheer, S. (2025). Enhanced LSTM-based AI model for accurate dissolved oxygen prediction in aquaculture systems. *Smart Agric. Technol.* 12:101140. doi: 10.1016/j.atech.2025.101140

Alnahit, A. O., Mishra, A. K., and Khan, A. A. (2022). Stream water quality prediction using boosted regression tree and random forest models. *Stoch. Environ. Res. Risk Assess.* 36, 2661–2680. doi: 10.1007/s00477-021-02152-4

Al-Sulttani, A. O., Al-Mukhtar, M., Roomi, A. B., Farooque, A. A., Khedher, K. M., and Yaseen, Z. M. (2021). Proposition of new ensemble data-intelligence models for surface water quality prediction. *IEEE Access* 9, 108527–108541. doi: 10.1109/ACCESS.2021.3100490

Azrour, M., Dargaoui, S., Mabrouki, J., Guezzaz, A., Benkirane, S., Shafik, W., et al. (2024). "A survey of machine and deep learning applications in the assessment of water quality," in *Technical and Technological Solutions Towards a Sustainable Society and Circular Economy*, (Cham: Springer Nature Switzerland), 471–483.

Basirian, S., Najafzadeh, M., and Demir, I. (2026). Water quality monitoring for coastal hypoxia: integration of satellite imagery and machine learning models. *Mar. Pollut. Bull.* 222:118735. doi: 10.1016/j.marpolbul.2025.118735

Bhardwaj, A., Dagar, V., Khan, M. O., Aggarwal, A., Alvarado, R., Kumar, M., et al. (2022). Smart IoT and machine learning-based framework for water quality assessment and device component monitoring. *Environ. Sci. Pollut. Res.* 29, 46018–46036. doi: 10.1007/s11356-022-19014-3

Bi, J., Lin, Y., Dong, Q., Yuan, H., and Zhou, M. (2021). Large-scale water quality prediction with integrated deep neural network. *Inf. Sci.* 571, 191–205. doi: 10.1016/j.ins.2021.04.057

Bi, J., Zhang, L., Yuan, H., and Zhang, J. (2023). Multi-indicator water quality prediction with attention-assisted bidirectional LSTM and encoder-decoder. *Inf. Sci.* 625, 65–80. doi: 10.1016/j.ins.2022.12.091

Cao, X., Ren, N., Tian, G., Fan, Y., and Duan, Q. (2021). A three-dimensional prediction method of dissolved oxygen in pond culture based on attention-GRU-GBRT. *Comput. Electron. Agric.* 181:105955. doi: 10.1016/j.compag.2020.105955

Chadalavada, S., Yaman, S., Sengur, A., Hafeez-Baig, A., Tan, R. S., Barua, P. D., et al. (2025). Gated LNN: gated liquid neural networks for accurate water quality index prediction and classification. *IEEE Access.* doi: 10.1109/ACCESS.2025.3561593

Chen, P., Wang, B., Wu, Y., Wang, Q., Huang, Z., and Wang, C. (2023). Urban river water quality monitoring based on self-optimizing machine learning method using multi-source remote sensing data. *Ecol. Indic.* 146:109750. doi: 10.1016/j.ecolind.2022.109750

Choudhary, R., Kumar, A., C, P., Naik, M. M., Choudhury, M., and Khan, N. A. (2025). Predicting water quality index using stacked ensemble regression and SHAP based explainable artificial intelligence. *Sci. Rep.* 15:31139. doi: 10.1038/s41598-025-09463-4

Dawood, T., Elwakil, E., Novoa, H. M., and Delgado, J. F. G. (2021). Toward urban sustainability and clean potable water: prediction of water quality via artificial neural networks. *J. Clean. Prod.* 291:125266. doi: 10.1016/j.jclepro.2020.125266

Deng, T., Chau, K. W., and Duan, H. F. (2021). Machine learning based marine water quality prediction for coastal hydro-environment management. *J. Environ. Manag.* 284:112051. doi: 10.1016/j.jenvman.2021.112051

Devagopal, A. M., Ashwin, V., Nidhin, S. N., Arjun Rathya, R., and Sn, J. (2022). *Prediction of water Quality Parameters of river periyar using Regression Models 2022 2nd International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE)*. IEEE, 53–57.

Dheda, D., Cheng, L., and Abu-Mahfouz, A. M. (2022). Long short term memory water quality predictive model discrepancy mitigation through genetic algorithm optimisation and ensemble modeling. *IEEE Access* 10, 24638–24658. doi: 10.1109/ACCESS.2022.3152818

Dilmi, S., and Ladjal, M. (2021). A novel approach for water quality classification based on the integration of deep learning and feature extraction techniques. *Chemometr. Intell. Lab. Syst.* 214:104329. doi: 10.1016/j.chemolab.2021.104329

DSouza, J. N., Santhosh, S., and Jins, D. (2023). *Predicting Quality of ground water using different deep Learning Models 2023 Fourth International Conference on Smart Technologies in Computing, Electrical and Electronics (ICSTCEE)*. Bengaluru: IEEE, 1–6.

Ehteram, M., Ahmed, A. N., Sherif, M., and El-Shafie, A. (2024). An advanced deep learning model for predicting water quality index. *Ecol. Indic.* 160:111806. doi: 10.1016/j.ecolind.2024.111806

El Moussaoui, Y., Adda, M., Lessard, L., Langlais, T., and Turcotte, S. (2025). RAG pipeline for private well contamination guidance: a comparative study of retrieval and generation strategies. *Procedia Comput. Sci.* 272, 335–342. doi: 10.1016/j.procs.2025.10.213

Elbeltagi, A., Pande, C. B., Kouadri, S., and Islam, A. R. M. T. (2022). Applications of various data-driven models for the prediction of groundwater quality index in the Akot basin, Maharashtra, India. *Environ. Sci. Pollut. Res.* 29, 17591–17605. doi: 10.1007/s11356-021-17064-7

Elmotawakkil, A., Enneya, N., Bhagat, S. K., Ouda, M. M., and Kumar, V. (2025). Advanced machine learning models for robust prediction of water quality index and classification. *J. Hydroinf.* 27, 299–319. doi: 10.2166/hydro.2025.290

Erickson, M. L., Elliott, S. M., Brown, C. J., Stackelberg, P. E., Ransom, K. M., Reddy, J. E., et al. (2021). Machine-learning predictions of high arsenic and high manganese at drinking water depths of the glacial aquifer system, northern continental United States. *Environ. Sci. Technol.* 55, 5791–5805. doi: 10.1109/ICSSAS57918.2023.10331622

Feng, D., Han, Q., Xu, L., Sohel, F., Hassan, S. G., and Liu, S. (2024). An ensembled method for predicting dissolved oxygen level in aquaculture environment. *Ecol. Inform.* 80:102501. doi: 10.1016/j.ecoinf.2024.102501

Gambín, Á. F., Angelats, E., González, J. S., Miozzo, M., and Dini, P. (2021). Sustainable marine ecosystems: deep learning for water quality assessment and forecasting. *IEEE Access* 9, 121344–121365. doi: 10.1109/ACCESS.2021.3109216

Gandhimathi, S. K., Muppagowni, G. K., Sandhya, K., Arava, R. R., and Kotapati, G. (2023). *A Proficient Prediction Mechanism for Analyzing Water Quality Using Machine Learning Algorithms 2023 International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS)*. Erode: IEEE, 552–559.

Grbčić, L., Družeta, S., Mauša, G., Lipić, T., Lušić, D. V., Alvir, M., et al. (2022). Coastal water quality prediction based on machine learning with feature interpretation and spatio-temporal analysis. *Environ. Model. Softw.* 155:105458. doi: 10.1016/j.envsoft.2022.105458

Huan, J., Li, H., Li, M., and Chen, B. (2020). Prediction of dissolved oxygen in aquaculture based on gradient boosting decision tree and long short-term memory network: a study of Chang Zhou fishery demonstration base, China. *Comput. Electron. Agric.* 175:105530. doi: 10.1016/j.compag.2020.105530

Huang, M., Hu, B. Q., Jiang, H., and Fang, B. W. (2023). A water quality prediction method based on k-nearest-neighbor probability rough sets and PSO-LSTM: M. Huang et al. *Appl. Intell.* 53, 31106–31128. doi: 10.1007/s10489-023-05024-2

Huang, J., Liu, S., Hassan, S. G., Xu, L., and Huang, C. (2021). A hybrid model for short-term dissolved oxygen content prediction. *Comput. Electron. Agric.* 186:106216. doi: 10.1016/j.compag.2021.106216

Huang, S., Xia, J., Wang, Y., Lei, J., and Wang, G. (2024). Water quality prediction based on sparse dataset using enhanced machine learning. *Environ. Sci. Ecotechnol.* 20:100402. doi: 10.1016/j.ese.2024.100402

Infant, S. S., Vickram, S., Saravanan, A., Muthu, C. M., and Yuarajan, D. (2025). Explainable artificial intelligence for sustainable urban water systems engineering. *Results Eng.* 25:104349. doi: 10.1016/j.rineng.2025.104349

Jamshidzadeh, Z., Latif, S. D., Ehteram, M., Sheikh Khozani, Z., Ahmed, A. N., Sherif, M., et al. (2024). An advanced hybrid deep learning model for predicting total dissolved solids and electrical conductivity (EC) in coastal aquifers. *Environ. Sci. Eur.* 36:20. doi: 10.1186/s12302-024-00850

Jiang, Y., Li, C., Sun, L., Guo, D., Zhang, Y., and Wang, W. (2021). A deep learning algorithm for multi-source data fusion to predict water quality of urban sewer networks. *J. Clean. Prod.* 318:128533. doi: 10.1016/j.jclepro.2021.128533

Jiang, Y., and Yan, F. (2022). Aquaculture prediction model based on improved water quality parameter data prediction algorithm under the background of big data. *J. Appl. Math.* 2022:2071360. doi: 10.1155/2022/2071360

Kalaivanan, K., Chitra, P. K. A., Ramesh, V., and Rajkumar, Y. (2026). FedWQP: a federated deep learning technique for water quality prediction. *Int. J. River Basin Manag.* 1-20, 1–20. doi: 10.1080/15715124.2026.2638863

Khan, M. S. I., Islam, N., Uddin, J., Islam, S., and Nasir, M. K. (2022). Water quality prediction and classification based on principal component regression and gradient boosting classifier approach. *J. King Saud Univ. Comput. Inf. Sci.* 34, 4773–4781. doi: 10.1016/j.jksuci.2021.06.003

Khullar, S., and Singh, N. (2022). Water quality assessment of a river using deep learning Bi-LSTM methodology: forecasting and validation. *Environ. Sci. Pollut. Res.* 29, 12875–12889. doi: 10.1007/s11356-021-13875-w

Kisi, O., Azad, A., Kashi, H., Saeedian, A., Hashemi, S. A. A., and Ghorbani, S. (2019). Modeling groundwater quality parameters using hybrid neuro-fuzzy methods. *Water Resour. Manag.* 33, 847–861. doi: 10.1007/s11269-018-2147-6

Kisi, O., and Parmar, K. S. (2016). Application of least square support vector machine and multivariate adaptive regression spline models in long term prediction of river water pollution. *J. Hydrol.* 534, 104–112. doi: 10.1016/j.jhydrol.2015.12.014

Kiyani, H. S., Yoo, K., Yoo, H., Rosen, G., Quon, H., Rosenbaum, A., et al. (2025). "Identifying Physiochemical Variables Important to LLM Prediction of Coliform Bacteria Presence Companion," in *Proceedings of the 16th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics*, (), 1–11.

Kolli, S., Ranjani, M., Kavitha, P., Daniel, D. A. P., and Chandramauli, A. (2023). *2023 International Conference on Computer Communication and Informatics (ICCCI)*. IEEE, 1–5.

Kouadri, S., Elbeltagi, A., Islam, A. R. M. T., and Kateb, S. (2021). Performance of machine learning methods in predicting water quality index based on irregular data set: application on Illizi region (Algerian southeast). *Appl Water Sci* 11:190. doi: 10.1007/s13201-021-01528-9

Kouadri, S., Pande, C. B., Panneerselvam, B., Moharir, K. N., and Elbeltagi, A. (2022). Prediction of irrigation groundwater quality parameters using ANN, LSTM, and MLR models. *Environ. Sci. Pollut. Res.* 29, 21067–21091. doi: 10.1007/s11356-021-17084-3

Li, W., Wei, Y., An, D., Jiao, Y., and Wei, Q. (2022). LSTM-TCN: dissolved oxygen prediction in aquaculture, based on combined model of long short-term memory network and temporal convolutional network. *Environ. Sci. Pollut. Res.* 29, 39545–39556. doi: 10.1007/s11356-022-18914-8

Liu, L., Ji, X., Duan, B., Li, Y., Xing, H., Wang, B., et al. (2025). A long-term prediction model for key water quality based on transformer with parallel attention mechanism and adaptive spectral enhancement. *Environ. Geochem. Health* 47:322. doi: 10.1007/s10653-025-02642-2

Liu, H., Yang, R., Duan, Z., and Wu, H. (2021). A hybrid neural network model for marine dissolved oxygen concentrations time-series forecasting based on multi-factor analysis and a multi-model ensemble. *Engineering* 7, 1751–1765. doi: 10.1016/j.eng.2020.10.023

Mahesh, N., Babu, J. J., Nithya, K., and Arunmozhi, S. A. (2024). Water quality prediction using LSTM with combined normalizer for efficient water management. *Desalin. Water Treat.* 317:100183. doi: 10.1016/j.dwt.2024.100183

Mamat, N., Hamzah, F. M., and Jaafar, O. (2021). Hybrid support vector regression model and K-fold cross validation for water quality index prediction in Langat River, Malaysia. *bioRxiv*:2021-02. doi: 10.1101/2021.02.15.431242

Miller, T., Durlik, I., Adrianna, K., Kisiel, A., Cembrowska-Lech, D., Spychalski, I., et al. (2023). Predictive modeling of urban lake water quality using machine learning: a 20-year study. *Appl. Sci.* 13:11217. doi: 10.3390/app132011217

Moeinzadeh, H., Jegakumaran, P., Yong, K. T., and Withana, A. (2023). Efficient water quality prediction by synthesizing seven heavy metal parameters using deep neural network. *J. Water Process Eng.* 56:104349. doi: 10.1016/j.jwpe.2023.104349

Mohd Zebaral Hoque, J., Ab Aziz, N. A., Alelyani, S., Mohana, M., and Hosain, M. (2022). Improving water quality index prediction using regression learning models. *Int. J. Environ. Res. Public Health* 19:13702. doi: 10.3390/ijerph192013702

Mokhtar, A., Elbeltagi, A., Gyasi-Agyei, Y., Al-Ansari, N., and Abdel-Fattah, M. K. (2022). Prediction of irrigation water quality indices based on machine learning and regression models. *Appl Water Sci* 12:76. doi: 10.1007/s13201-022-01590-x

Mu, Z., Jiang, H., Zhao, J., Li, D., Gong, Z., and Xue, S. (2026). Physics-guided transformer-based modeling for surface water quality prediction in aquatic environments. *J. Water Process Eng.* 86:109908. doi: 10.1016/j.jwpe.2026.109908

Najafzadeh, M., Ahmadi-Rad, E. S., and Gebler, D. (2024). Ecological states of watercourses regarding water quality parameters and hydromorphological parameters: deriving empirical equations by machine learning models. *Stoch. Environ. Res. Risk Assess.* 38, 665–688. doi: 10.1007/s00477-023-02593-z

Najafzadeh, M., and Basirian, S. (2023). Evaluation of river water quality index using remote sensing and artificial intelligence models. *Remote Sens.* 15:2359. doi: 10.3390/rs15092359

Najafzadeh, M., and Ghaemi, A. (2019). Prediction of the five-day biochemical oxygen demand and chemical oxygen demand in natural streams using machine learning methods. *Environ. Monit. Assess.* 191:380. doi: 10.1007/s10661-019-7446-8

Najafzadeh, M., Ghaemi, A., and Emamgholizadeh, S. (2019). Prediction of wat quality parameters using evolutionary computing-based formulations. *Int. J. Environ. Sci. Technol.* 16, 6377–6396. doi: 10.1007/s13762-018-2049-4

Najafzadeh, M., and Niazmardi, S. (2021). A novel multiple-kernel support vector regression algorithm for estimation of water quality parameters. *Nat. Resour. Res.* 30, 3761–3775. doi: 10.1007/s11053-021-09895-5

Najah, A., Teo, F. Y., Chow, M. F., Huang, Y. F., Latif, S. D., Abdullah, S., et al. (2021). Surface water quality status and prediction during movement control operation order under COVID-19 pandemic: case studies in Malaysia. *Int. J. Environ. Sci. Technol.* 18, 1009–1018. doi: 10.1007/s13762-021-03139-y

Nallakaruppan, M. K., Gangadevi, E., Shri, M. L., Balusamy, B., Bhattacharya, S., and Selvarajan, S. (2024). Reliable water quality prediction and parametric analysis using explainable AI models. *Sci. Rep.* 14:7520. doi: 10.1038/s41598-024-56775-y

Nasir, N., Kansal, A., Alshaltone, O., Barneih, F., Sameer, M., Shanableh, A., et al. (2022). Water quality classification using machine learning algorithms. *J. Water Process Eng.* 48:102920. doi: 10.1016/j.jwpe.2022.102920

Nishat, M. H., Khan, M. H. R. B., Ahmed, T., Hossain, S. N., Ahsan, A., El-Sergany, M. M., et al. (2025). Comparative analysis of machine learning models for predicting water quality index in Dhaka's rivers of Bangladesh. *Environ. Sci. Eur.* 37:31. doi: 10.1186/s12302-025-01078-w

Ooi, K. S., Chen, Z., Poh, P. E., and Cui, J. (2022). BOD5 prediction using machine learning methods. *Water Supply* 22, 1168–1183. doi: 10.2166/ws.2021.202

Pandya, J. K., Khandelwal, S. S., Tipu, R. K., and Pandya, K. S. (2025). Advancing water quality management: an integrated approach using ensemble machine learning and real-time interactive visualization. *IEEE Access* 13, 92406–92428. doi: 10.1109/ACCESS.2025.3573589

Patel, J., Amipara, C., Ahanger, T. A., Ladhva, K., Gupta, R. K., Alsaab, H. O., et al. (2022). A machine learning-based water potability prediction model by using synthetic minority oversampling technique and explainable AI. *Comput. Intell. Neurosci.* 2022:9283293. doi: 10.1155/2022/9283293

Prasad, D. V. V., Venkataramana, L. Y., Kumar, P. S., Prasannamedha, G., Soumya, K., and Poornema, A. J. (2021). Prediction on water quality of a lake in Chennai, India using machine learning algorithms. *Desalin. Water Treat.* 218, 44–51. doi: 10.5004/dwt.2021.26970

Rahaman, M. H., Sajjad, H., Hussain, S., Masroor, M., and Sharma, A. (2024). Surface water quality prediction in the lower Thoubal river watershed, India: a hyper-tuned machine learning approach and DNN-based sensitivity analysis. *J. Environ. Chem. Eng.* 12:112915. doi: 10.1016/j.jece.2024.112915

Raheja, H., Goel, A., and Pal, M. (2022). Prediction of groundwater quality indices using machine learning algorithms. *Water Pract. Technol.* 17, 336–351. doi: 10.2166/wpt.2021.120

Ransom, K. M., Nolan, B. T., Stackelberg, P. E., Belitz, K., and Fram, M. S. (2022). Machine learning predictions of nitrate in groundwater used for drinking supply in the conterminous United States. *Sci. Total Environ.* 807:151065. doi: 10.1016/j.scitotenv.2021.151065

Rathinam, G. (2025). Quantized GAN-based CNN hybrid model with enhanced explainability for water quality assessment. *Int. j. inf. tecnol.* doi: 10.1007/s41870-025-02845-x

Sakaa, B., Elbeltagi, A., Boudibi, S., Chaffaï, H., Islam, A. R. M. T., Kulimushi, L. C., et al. (2022). Water quality index modeling using random forest and improved SMO algorithm for support vector machine in Saf-Saf river basin. *Environ. Sci. Pollut. Res.* 29, 48491–48508. doi: 10.1007/s11356-022-18644-x

Samuel, D. J., Sermet, Y., Cwiertny, D., and Demir, I. (2024). Integrating vision-based AI and large language models for real-time water pollution surveillance. *Water Environ. Res.* 96:e11092. doi: 10.1002/wer.11092

Satish, N., Anmala, J., Rajitha, K., and Varma, M. R. (2024). A stacking ANN ensemble model of ML models for stream water quality prediction of Godavari River basin, India. *Ecol. Inform.* 80:102500. doi: 10.1016/j.ecoinf.2024.102500

Shams, M. Y., Elshewey, A. M., El-Kenawy, E. S. M., Ibrahim, A., Talaat, F. M., and Tarek, Z. (2024). Water quality prediction using machine learning models based on grid search method. *Multimed. Tools Appl.* 83, 35307–35334. doi: 10.1007/s11042-023-16737-4

Shin, J., Yoon, S., Park, N. S., and Kim, Y. (2025). Development of water quality prediction model using LTSF-linear and complete ensemble empirical mode decomposition. *Desalin. Water Treat.* 323:101254. doi: 10.1016/j.dwt.2025.101254

Slatnia, A., Ouali, M. A., Ladjal, M., and Bennacer, H. (2026). Prediction and assessment of water quality index for surface water using 1D CNN and SVR models. *Environ. Sci. Pollut. Res.* 33, 5027–5056. doi: 10.1007/s11356-026-37556-8

Talukdar, S., Ahmed, S., Naikoo, M. W., Rahman, A., Mallik, S., Ningthoujam, S., et al. (2023). Predicting lake water quality index with sensitivity-uncertainty analysis using deep learning algorithms. *J. Clean. Prod.* 406:136885. doi: 10.1016/j.jclepro.2023.136885

Tejaswi, T., Manoj, C., Naidu, P. V. D., Santhosh, T., Akhil, P. V. S., and Ganesan, V. (2022). *Nexus of Water Quality Prediction by ANN 2022 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES)*. Chennai: IEEE, 1–5.

Tiyasha, Tung, T. M., and Yaseen, Z. M. (2021). Deep learning for prediction of water quality index classification: tropical catchment environmental assessment. *Nat. Resour. Res.* 30, 4235–4254. doi: 10.1007/s11053-021-09922-5

Uddin, M. G. (2023). *Development of a novel water Quality index model using data Science Approaches (Doctoral dissertation)*. NUI Galway.

Uddin, M. G., Nash, S., Diganta, M. T. M., Rahman, A., and Olbert, A. I. (2022). Robust machine learning algorithms for predicting coastal water quality index. *J. Environ. Manag.* 321:115923. doi: 10.1016/j.jenvman.2022.115923

Uddin, M. G., Nash, S., Rahman, A., and Olbert, A. I. (2023). A novel approach for estimating and predicting uncertainty in water quality index model using machine learning approaches. *Water Res.* 229:119422. doi: 10.1016/j.watres.2022.119422

Vellingiri, J., Kalaivanan, K., Gopinath, M. P., and Gobinath, C. (2023). Strategies for classifying water quality in the Cauvery River using a federated learning technique. *Int. J. Cogn. Comput. Eng.* 4, 187–193. doi: 10.1016/j.ijcce.2023.04.004

Vijay Anand, M., Sohitha, C., Saraswathi, G. N., and Lavanya, G. V. (2023). Water quality prediction using CNN. *J. Phys. Conf. Ser.* 2484:012051.

Vijay, S., and Kamaraj, K. (2021). Prediction of water quality index in drinking water distribution system using activation functions based Ann. *Water Resour. Manag.* 35, 535–553. doi: 10.1007/s11269-020-02729-8

Wang, Z., Duan, L., Shuai, D., and Qiu, T. (2024). Research on water environmental indicators prediction method based on EEMD decomposition with CNN-BiLSTM. *Sci. Rep.* 14:1676. doi: 10.1038/s41598-024-51936-5

Wang, Y. N., Li, B., and Yang, G. (2023). Stream water quality optimized prediction based on human activity intensity and landscape metrics with regional heterogeneity in Taihu Basin, China. *Environ. Sci. Pollut. Res. Int.* 30, 4986–5004. doi: 10.1007/s11356-022-22536-5

Wang, Z., Wang, Q., and Wu, T. (2023). A novel hybrid model for water quality prediction based on VMD and IGOA optimized for LSTM. *Front. Environ. Sci. Eng.* 17:88. doi: 10.1007/s11783-023-1688-y

William, P., Oyebode, O. J., Ramu, G., Gupta, M., Bordoloi, D., and Shrivastava, A. (2023). *Artificial Intelligence based Models to Support water Quality Prediction using Machine Learning Approach 2023 International Conference on Circuit Power and Computing Technologies (ICCPCT)*. Kollam: IEEE, 1496–1501.

Xue, J., Yuan, C., Ji, X., and Zhang, M. (2024). Predictive modeling of nitrogen and phosphorus concentrations in rivers using a machine learning framework: a case study in an urban-rural transitional area in Wenzhou China. *Sci. Total Environ.* 910:168521. doi: 10.1016/j.scitotenv.2023.168521

Yadav, P., Upadhyay, S., Singh, R., Sharma, P., Patil, U., churi, M., et al. (2025). Design and development of a machine learning-based decision support system for water quality prediction in aquaponic farming. *Int. J. Adv. Comput. Eng. Commun. Technol.* 14, 130–135. doi: 10.65521/ijacect.v14i1.333

Yan, K., Li, C., Zhao, R., Zhang, Y., Duan, H., and Wang, W. (2023). Predicting the ammonia nitrogen of wastewater treatment plant influent via integrated model based on rolling decomposition method and deep learning algorithm. *Sustain. Cities Soc.* 94:104541. doi: 10.1016/j.scs.2023.104541

Yang, J., Jia, L., Guo, Z., Shen, Y., Li, X., Mou, Z., et al. (2023). Prediction and control of water quality in recirculating aquaculture system based on hybrid neural network. *Eng. Appl. Artif. Intell.* 121:106002. doi: 10.1016/j.engappai.2023.106002

Yang, Y., Xiong, Q., Wu, C., Zou, Q., Yu, Y., Yi, H., et al. (2021). A study on water quality prediction by a hybrid CNN-LSTM model with attention mechanism. *Environ. Sci. Pollut. Res.* 28, 55129–55139. doi: 10.1007/s11356-021-14687-8

Zhang, Z., Huang, J., Duan, S., Huang, Y., Cai, J., and Bian, J. (2022). Use of interpretable machine learning to identify the factors influencing the nonlinear linkage between land use and river water quality in the Chesapeake Bay watershed. *Ecol. Indic.* 140:108977. doi: 10.1016/j.ecolind.2022.108977

Zhang, L., Jiang, Z., He, S., Duan, J., Wang, P., and Zhou, T. (2022). Study on water quality prediction of urban reservoir by coupled CEEMDAN decomposition and LSTM neural network model. *Water Resour. Manag.* 36, 3715–3735. doi: 10.1007/s11269-022-03224-y

Zhang, X., and Li, D. (2023). Multi-input multi-output temporal convolutional network for predicting the long-term water quality of ocean ranches. *Environ. Sci. Pollut. Res.* 30, 7914–7929. doi: 10.1007/s11356-022-22588-7

Zhang, Y., Li, C., Jiang, Y., Sun, L., Zhao, R., Yan, K., et al. (2022). Accurate prediction of water quality in urban drainage network with integrated EMD-LSTM model. *J. Clean. Prod.* 354:131724. doi: 10.1016/j.jclepro.2022.131724

Zhang, Y., Yao, X., Wu, Q., Huang, Y., Zhou, Z., Yang, J., et al. (2021). Turbidity prediction of lake-type raw water using random forest model based on meteorological data: a case study of tai lake, China. *J. Environ. Manag.* 290:112657. doi: 10.1016/j.jenvman.2021.112657

Zhong, H., Yuan, Y., Luo, L., Ye, J., Chen, M., and Zhong, C. (2022). Water quality prediction of MBR based on machine learning: a novel dataset contribution analysis method. *J. Water Process Eng.* 50:103296. doi: 10.1016/j.jwpe.2022.103296

Zounemat-Kermani, M., Keshtegar, B., Kisi, O., and Scholz, M. (2021). Towards a comprehensive assessment of statistical versus soft computing models in hydrology: application to monthly pan evaporation prediction. *Water* 13:2451. doi: 10.3390/w13172451